THERMODYNAMICS AND THERMAL SYSTEMS (BAMEE203)

VIT®
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Syllabus

Module: 1	Laws of Thermodynamics	9 hours
Basic concepts; work and heat transfer; Zeroth law of thermodynamics; first law of thermodynamics - internal energy, working fluid - ideal gas; energy analysis of closed and open systems; second law of thermodynamics - heat engines, heat pumps, and refrigerators; reversible and irreversible processes; entropy; introduction to exergy		
Module: 2	**Gas and Vapour Power Cycles**	**9 hours**
Air standard cycles - Carnot, Otto, Diesel, Brayton cycles; properties of pure substances; property diagrams for phase change processes; vapor power cycles - Rankine cycle, reheat Rankine cycle, and regenerative Rankine cycle.		
Module: 3	**IC Engines**	**8 hours**
Introduction to IC engines; classification; testing and performance; simple carburetor; ignition system; fuel injection system - MPFI, CRDI, GDI; supercharging and turbocharging; combustion; knocking and detonation.		
Module: 4	**Air Compressor and Steam Nozzles**	**8 hours**
Reciprocating compressors - construction, working, effect of clearance volume, volumetric efficiency, isothermal efficiency, multi-staging; steam nozzles - one-dimensional steady flow through convergent-divergent nozzle; metastable flow.		
Module: 5	**Refrigeration and Air Conditioning**	**9 hours**
Vapor compression refrigeration system - components, working, P-h and T-s diagrams, calculation of COP, effect of sub-cooling and superheating, selection and properties of refrigerants; vapor absorption system; psychrometry - properties, processes, and chart; heating and cooling load calculations; air-conditioning systems.		

Course Objectives and Outcomes

Course Objectives

1. To explain the basic concepts of thermodynamics, including work, heat transfer, and energy analysis of systems.
2. To teach gas and vapor power cycles, IC engines, compressors, and nozzles for practical energy applications.
3. To guide the understanding of refrigeration, air-conditioning, and modern energy systems for evaluating performance and efficiency.

Course Outcomes (COs)

1. Apply the laws of thermodynamics to solve energy-related problems in closed and open systems.
2. Analyze gas and vapor power cycles to determine efficiency and performance
3. Calculate the performance of IC engines, compressors, and nozzles using standard principles
4. Apply refrigeration and air-conditioning principles to calculate system performance and design parameters
5. Assess contemporary energy systems and related technologies for practical engineering applications.

Text and Reference Books

Text-Books

1. Yunus A. Cengel, Michael A. Boles and Mehmet Kanoglu, Thermodynamics: An Engineering Approach, 2019, 9th Edition, McGraw Hill Education.
2. Nag P. K., Engineering Thermodynamics, 2017, 6th Edition, McGraw Hill Education.
3. Rajput R.K, "Thermal Engineering", Laxmi Publications Pvt Ltd, 10th Edition, 2017

Reference Books

1. Eastop T. D, McConkey A, "Applied Thermodynamics for Engineering Technologists", Pearson Education Ltd, 5th Edition, 2009
2. Ganesan, V., "Internal combustion engines", McGraw Hill Education Pvt Ltd, 1st Edition, 2012
3. Manohar Prasad, "Refrigeration and Air Conditioning", New Age International, 3rd Edition, 2015

Evaluation Method & Assignment Details

Component	Coverage	Weightage	Relevant CO's	Max. Marks
CAT - 1	Module 1-2	15%	CO1, CO2	50
CAT - 2	Module 2, 4 (Open Notes)	15%	CO2, CO4	50
Assignments	DA / Quiz / Project Review	30%	CO1 – CO5	30
FAT	2 Questions from each module.	40%	CO1-CO5	100
Total		**100%**		**230**

Assignment	Max. Marks
Assignment 1 (DA1)	10
Assignment 2 (DA2)	10
Assignment 3 (DA3)	10
Assignments (DA)	**30**

Module

4

Air Compressor and Steam Nozzles

CO-3

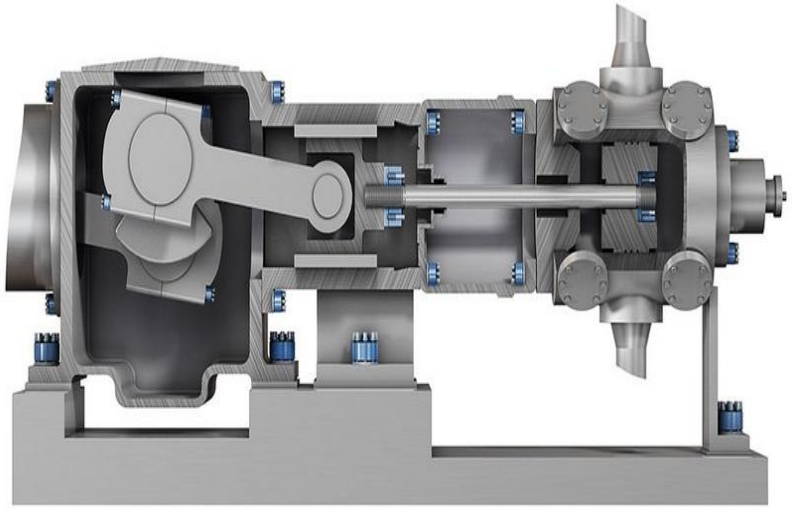
Air Compressor and Steam Nozzles 8 hours

- Reciprocating compressors - construction, working, effect of clearance volume, volumetric efficiency, isothermal efficiency, multi-staging;
- steam nozzles - one-dimensional steady flow through convergent-divergent nozzle;
- metastable flow.

COMPRESSOR

COMPRESSOR – A device which takes a definite quantity of fluid (usually gas, and most often air) and deliver it at a required pressure.





Classification of Compressor

Compressors can be classified based on the working principle (positive displacement or dynamic) and further into various types as shown below.

COMPRESSOR TYPES

POSITIVE DISPLACEMENT

Traps a fixed volume of gas and reduces its volume to increase pressure.

RECIPROCATING

Uses a piston moving back and forth in a cylinder.



SINGLE-ACTING

Compression occurs on one side of the piston.



DOUBLE-ACTING

Compression occurs on both sides of the piston.



DIAPHRAGM

Uses a flexible diaphragm to compress the gas.

ROTARY

Uses rotating elements to trap and compress gas.



LOBE

Two lobes rotate in opposite directions.



LIQUID RING

Uses a liquid ring as a seal and compression medium.



SCREW

Two helical screws rotate to compress the gas.



VANE

Sliding vanes trap and compress the gas.

DYNAMIC

Continuously accelerates the gas and converts velocity into pressure.

CENTRIFUGAL

Increases pressure by centrifugal force using a rotating impeller.



- High flow rate
- Moderate pressure rise
- Suitable for continuous operation

AXIAL

Increases pressure by a series of rotating and stationary blades (along the axis).



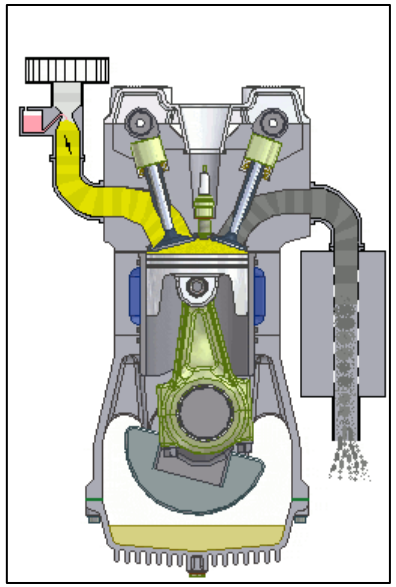
- Very high flow rate
- Low to moderate pressure rise (per stage)
- Common in gas turbines and jet engines

In all types, the work input to the compressor increases the pressure (and temperature) of the gas for its delivery to a higher-pressure system.

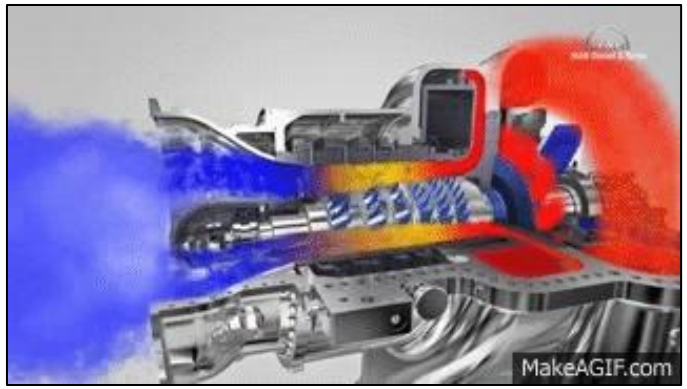
Classification of compressors



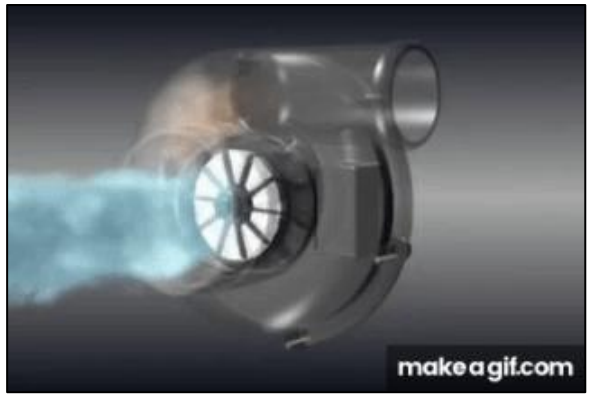
Rotary Screw Compressors



Reciprocating Air Compressors



Axial Compressors



Centrifugal Compressors

Reciprocating Compressors

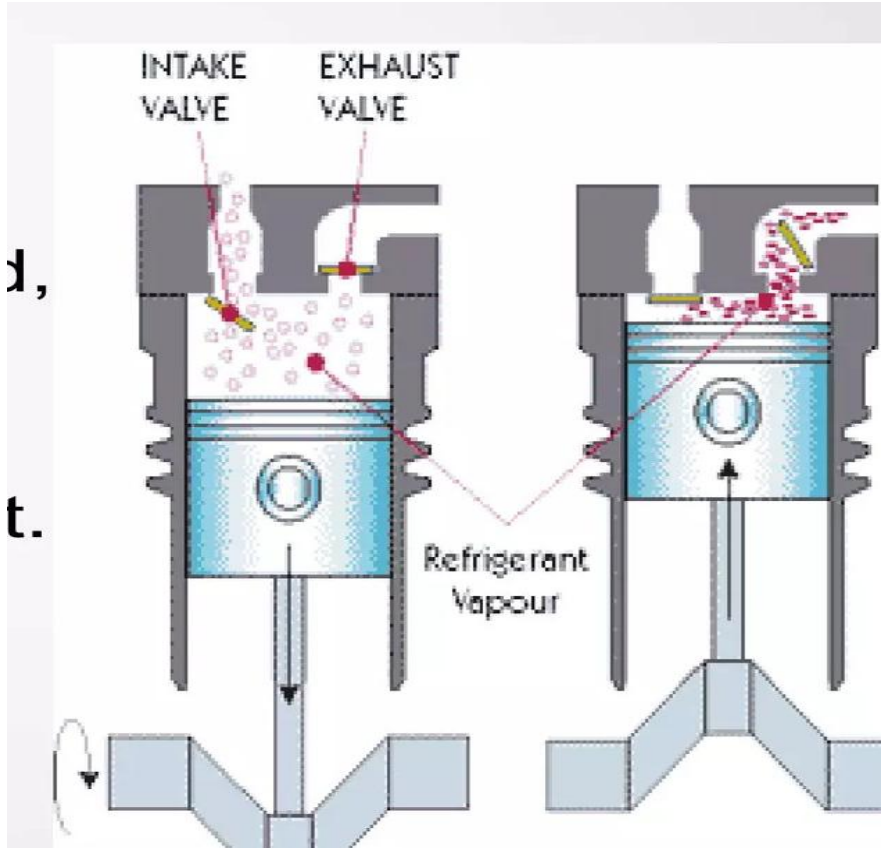
Working sequence

A positive-displacement compressor using pistons driven by a crankshaft.

Intake gas enters through the suction manifold into the cylinder.

The piston moves in reciprocating motion and compresses the gas.

The compressed gas is discharged at higher pressure.



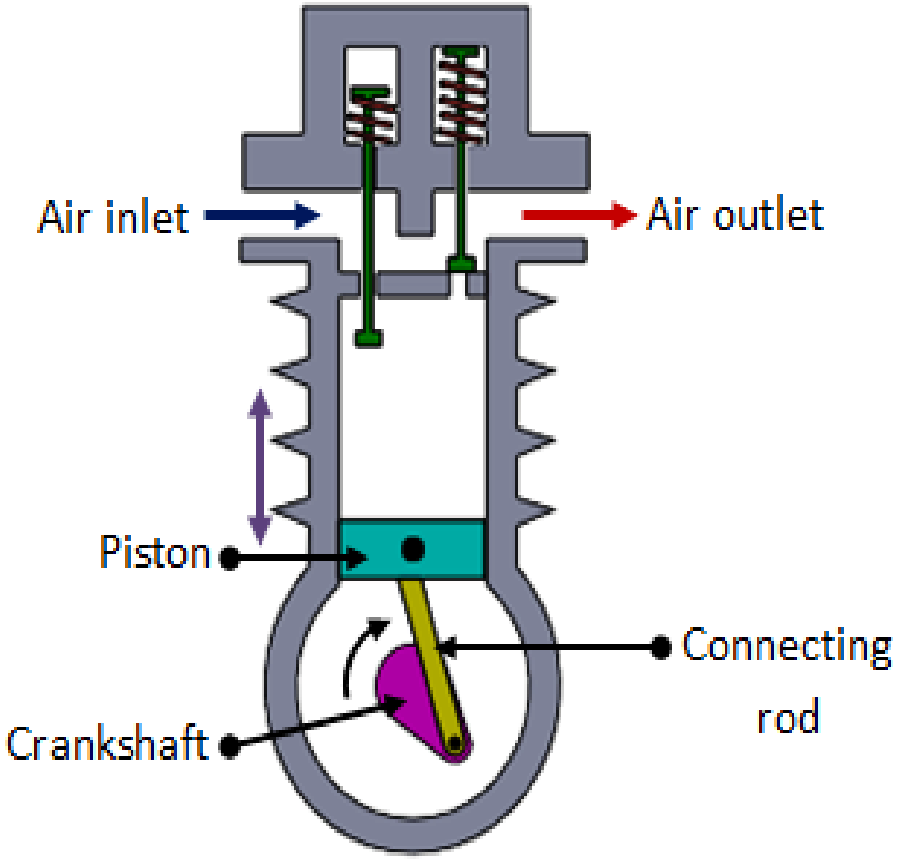
Classification of Reciprocating compressor

A) Depending on the operation
   1) Single acting compressor.
   2) Double acting compressor.

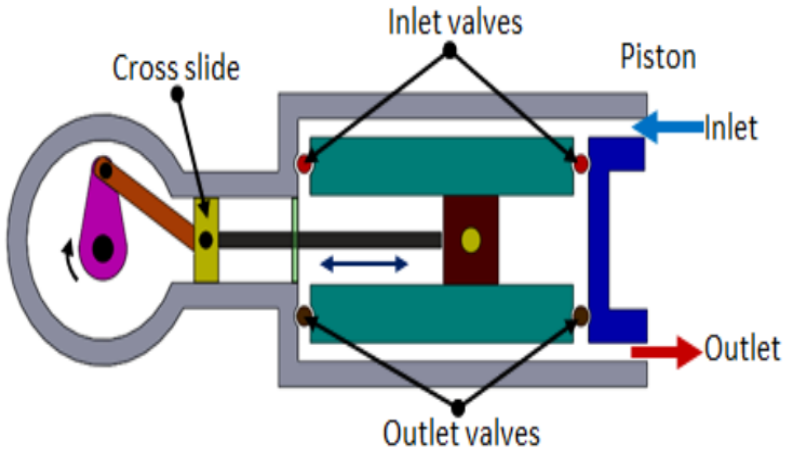
B) Depending on the number of stages
   1) Single Stage.
   2) Multi Stage.

C) Method of cooling
   1) Air cooled compressors.
   2) Water cooled compressors.

Single acting compressor



Double Acting Compressor

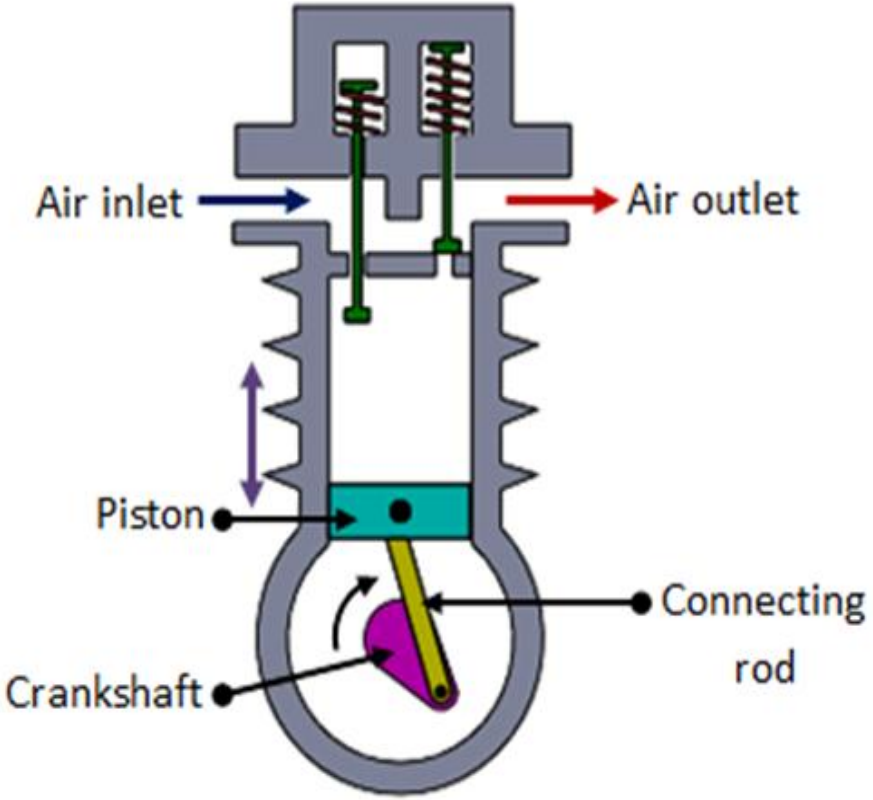


Delivery happens in both directions of piston motion

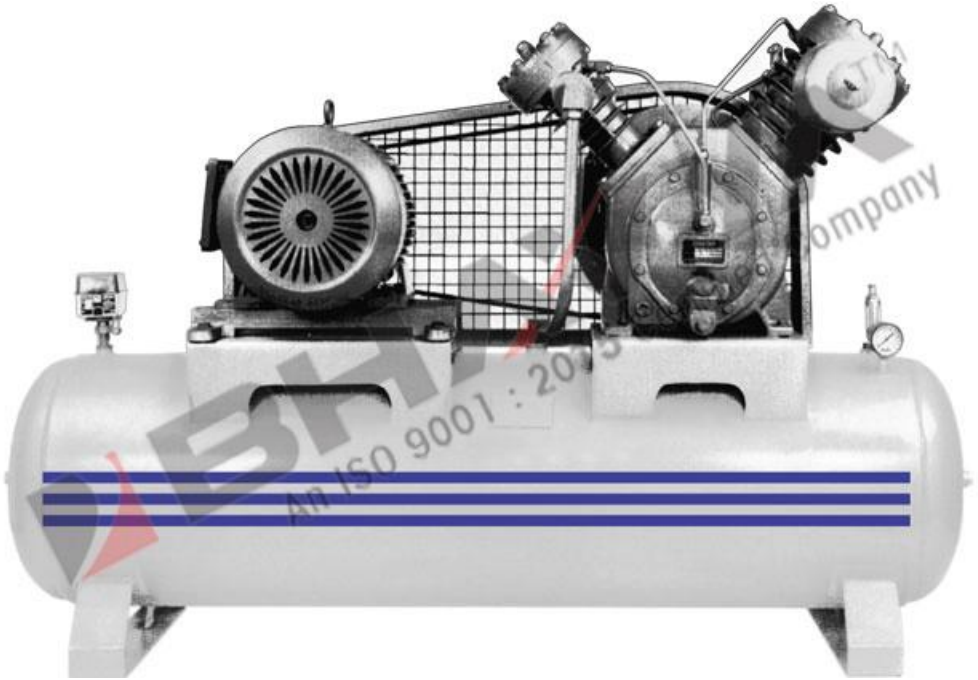
https://www.youtube.com/watch?v=E6_jw841vKE&t=14s

- A "single acting reciprocating compressor" only compresses air on one stroke of the piston (typically the upstroke), while a "double acting reciprocating compressor" compresses air on both the upstroke and down stroke of the piston

Single Stage air Compressor



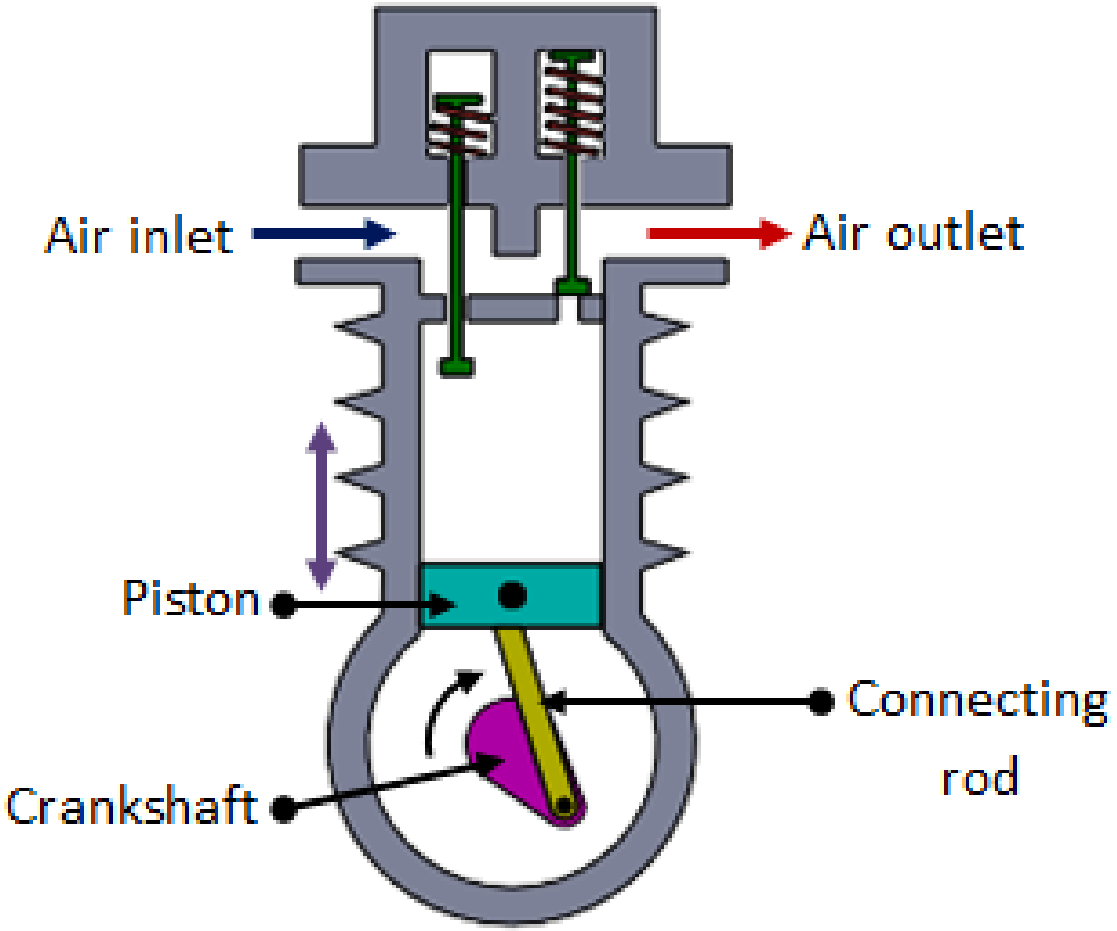
Multi Stage air compressor



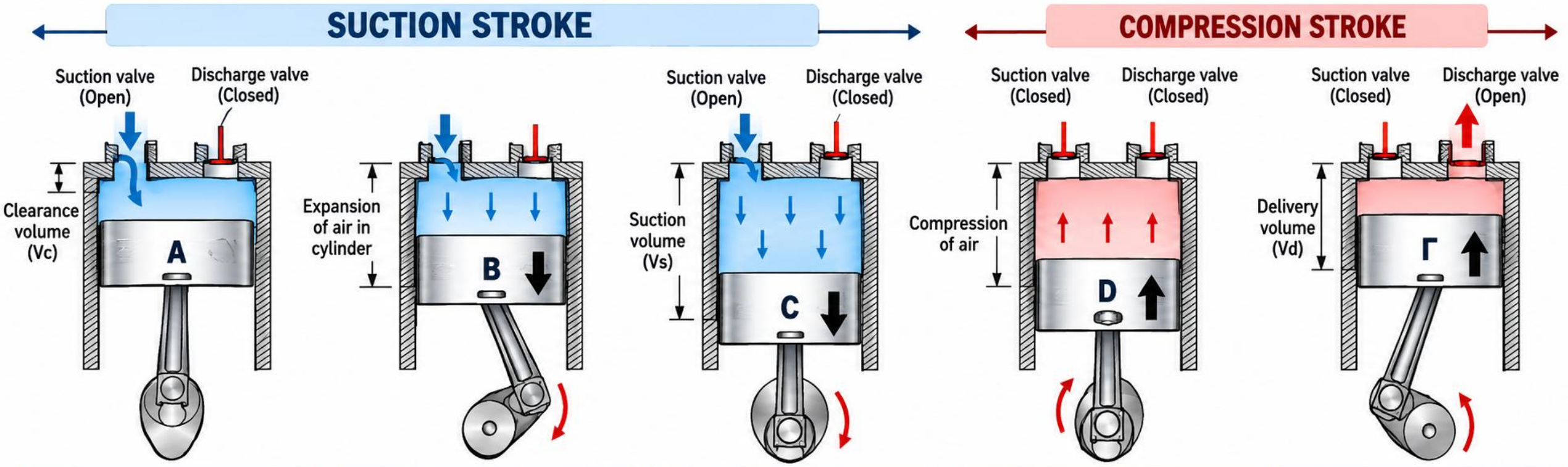
In a multistage air compressor, one cylinder is smaller than the others because it is designed to handle the already compressed air from the previous stage, which means it needs a smaller volume to achieve the desired high pressure.

Positive Displacement Compressor

Positive displacement compressors causes movement by trapping a fixed amount of air then forcing (displacing) that trapped volume into the discharge pipe.



Working of a Reciprocating Compressor



A. Start of suction stroke
(Piston at Top Dead Center)

Suction valve opens due to lower pressure in cylinder. Air enters the cylinder.

B. Suction stroke
(Piston moving down)

Suction valve remains open. Cylinder volume increases and air is drawn in at nearly constant pressure (≈ atmospheric).

C. End of suction stroke
(Piston at Bottom Dead Center)

Maximum volume is reached. Suction valve closes as piston starts moving upward.

D. Compression stroke
(Piston moving up)

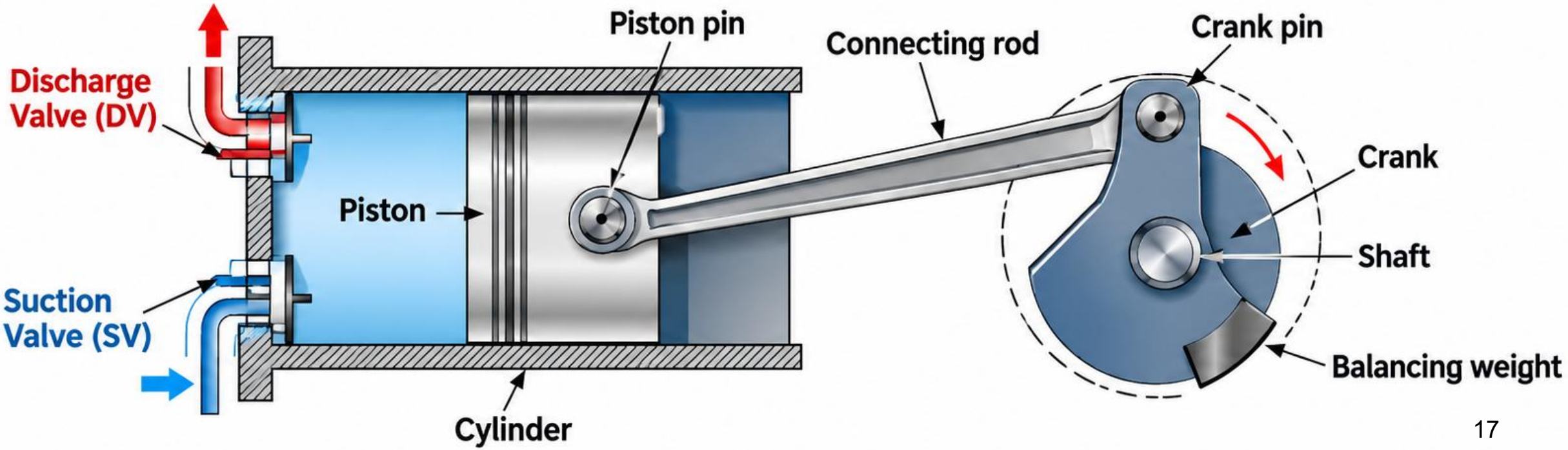
Both valves are closed. Air is compressed, so the pressure and temperature rise.

E. End of compression stroke
(Piston at Top Dead Center)

When cylinder pressure exceeds delivery pressure, discharge valve opens and compressed air leaves the cylinder.

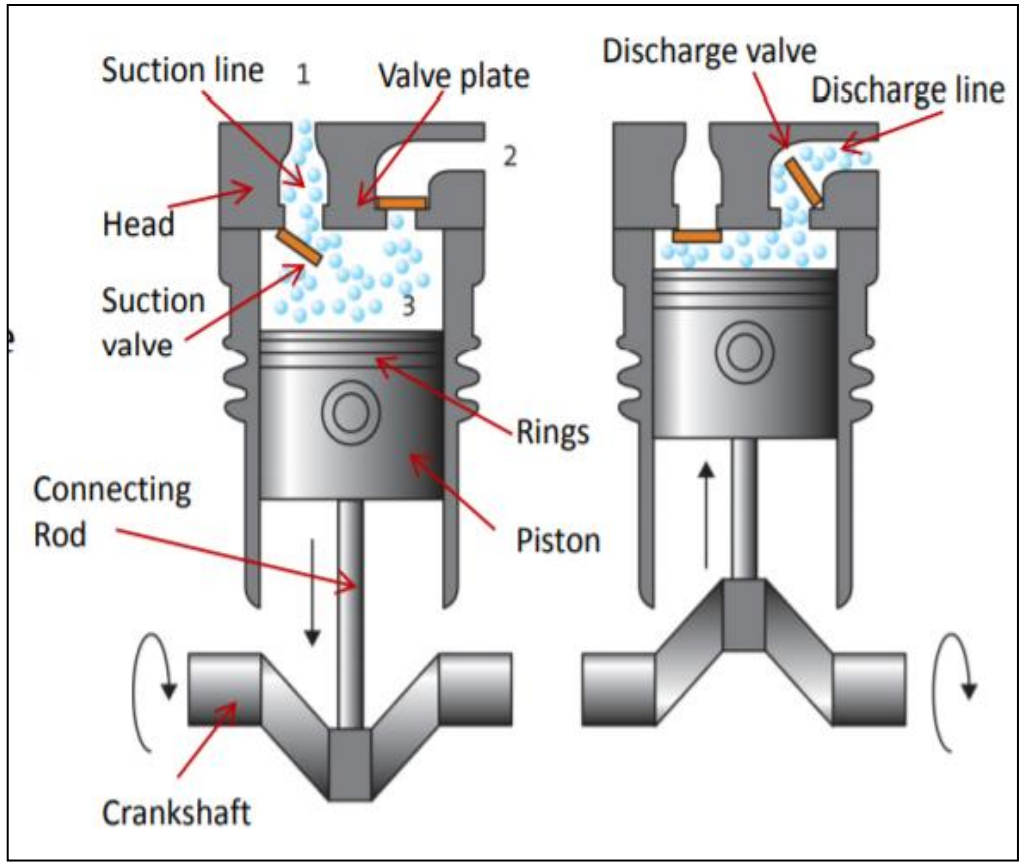
Construction and Working

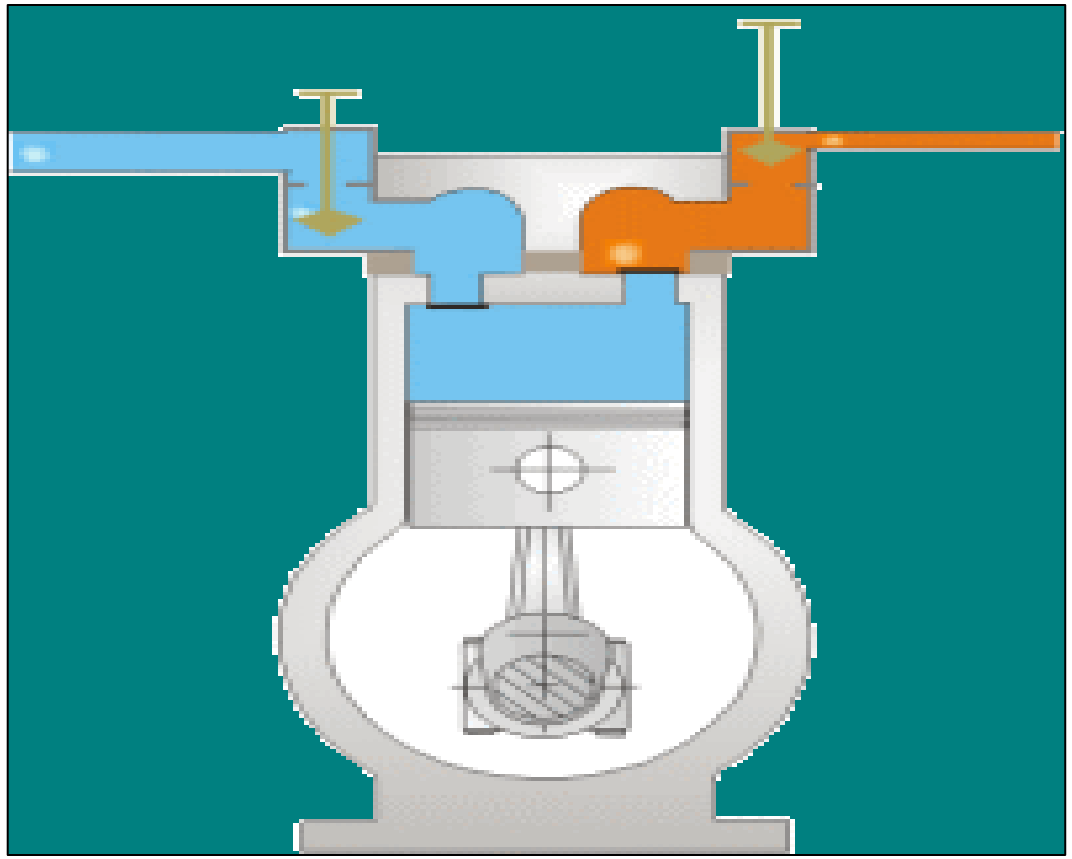
- Construction is **similar to that of an IC engine** – piston reciprocates within the cylinder. Has connecting rod, crankshaft etc.
- Operation is **similar to a non-firing engine** – suction occurs during downstroke, compression + delivery in upstroke – one crank revolution.
- Valves are opened and closed by the **Δp** between manifold and cylinder.
- Piston can be **single acting or double acting** (compression on both sides).



Reciprocating Compressor

- Uses pistons driven by a crankshaft to deliver gases at high pressure.
- The intake gas enters the suction manifold, then flows into the compression cylinder.
- It gets compressed by a piston driven in a reciprocating motion via a crankshaft.
- Discharged at higher pressure





Reciprocating Compressor

Compression ratio:- It is defined as the ratio of volume of air before compression to the volume of air after compression.

Compressor capacity:- It is the quantity of air actually delivered by a compressor in m^3/minute or m^3/sec.

Free air Delivered (FAD):- It is the volume of compressed air delivered by a compressor, expressed in terms of its volume at the specified free-air inlet conditions (i.e. temperature and pressure). FAD is *not the actual volume* of air at the high discharge pressure; it is the *equivalent volume at the reference inlet conditions.*

Swept Volume:- The volume displaced or swept by piston when it moves between top dead center and bottom dead center.

Clearance volume:- It is the difference between the total volume and the swept volume, basically the gap that remains between the piston head and the cylinder head when at top dead center

Reciprocating Compressor

Efficiency Indices

- Isothermal efficiency compares the actual work with isothermal work

$$\eta_T = \frac{\text{Isothermal work}}{\text{Actual work}}$$

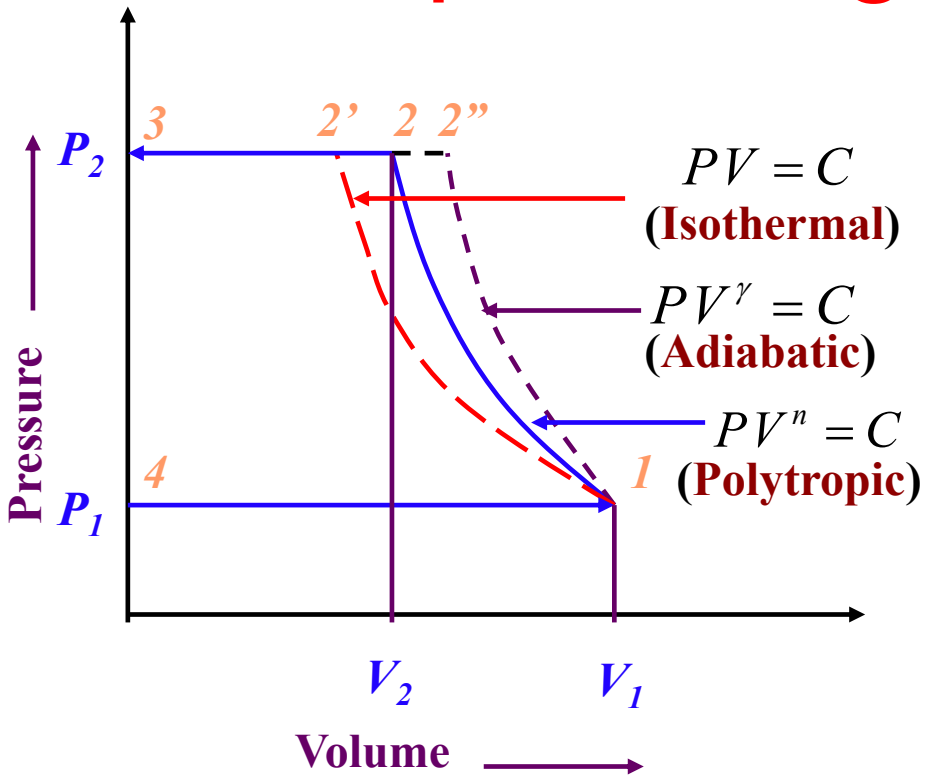
- Isentropic efficiency compares actual work to isentropic work

$$\eta_s = \frac{\text{Isentropic work}}{\text{Actual work}}$$

- Volumetric efficiency is the ratio of actual volume of gas taken in per cycle to the ideal volume (displacement/swept/stroke volume)

$$\eta_v = \frac{V_{actual}}{V_d} = \frac{m_{actual}}{m_{ideal}}$$

Reciprocating Compressor



4 – 1 : Volume V_1 of air aspirated into Compressor, at P_1 and T_1.

1 – 2 : Air compressed according to $PV^n = Const.$ from P_1 to P_2.

$\longrightarrow$ Temp increase from T_1 to T_2.

2 – 3 : Compressed air at P_2 and V_2 with temperature T_2 is delivered.

Reciprocating Compressor

During Compression, due to the excess temperature above surrounding, the air will exchange the heat to the surrounding.

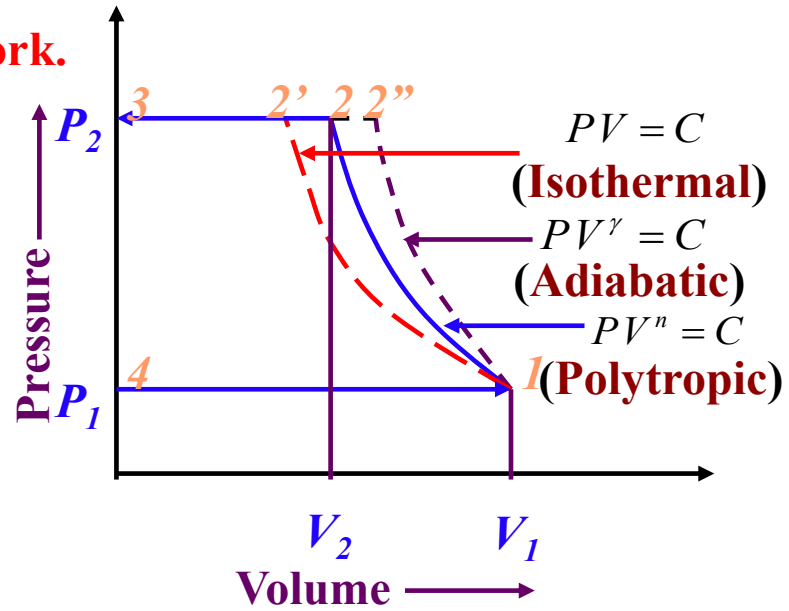
As Compressor is a work consuming device, every effort is desired to reduce the work.

Work done = Area under *P-V* curve

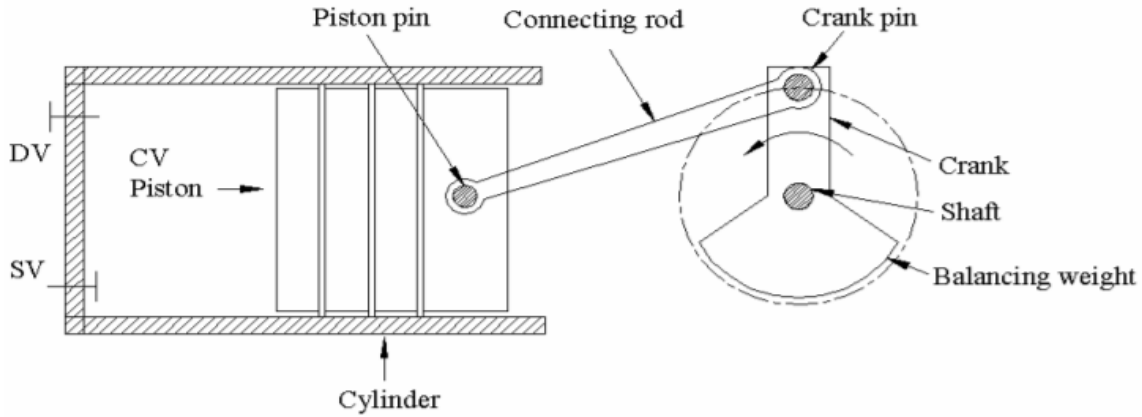
$\Rightarrow$ **1 – 2" : Adiabatic Compression = Max. Work.**

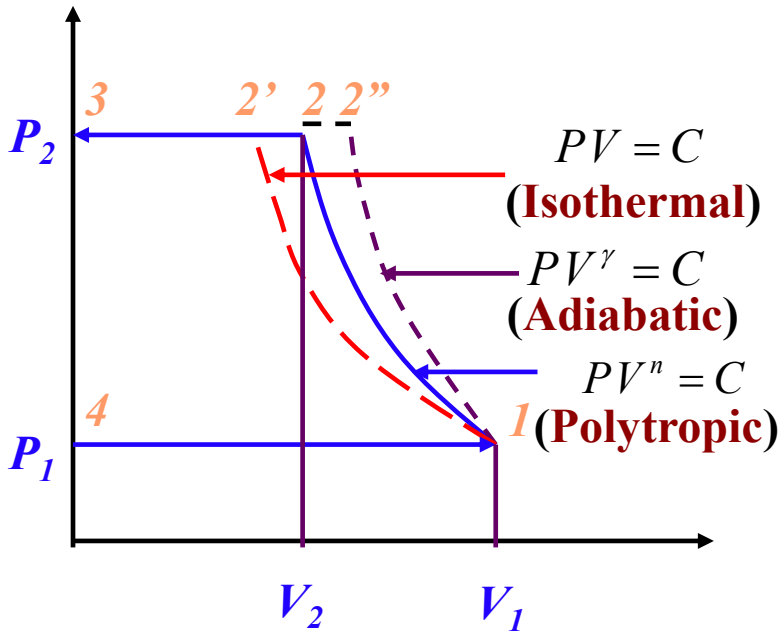
$\Rightarrow$ **1 – 2 : Polytropic Compression**

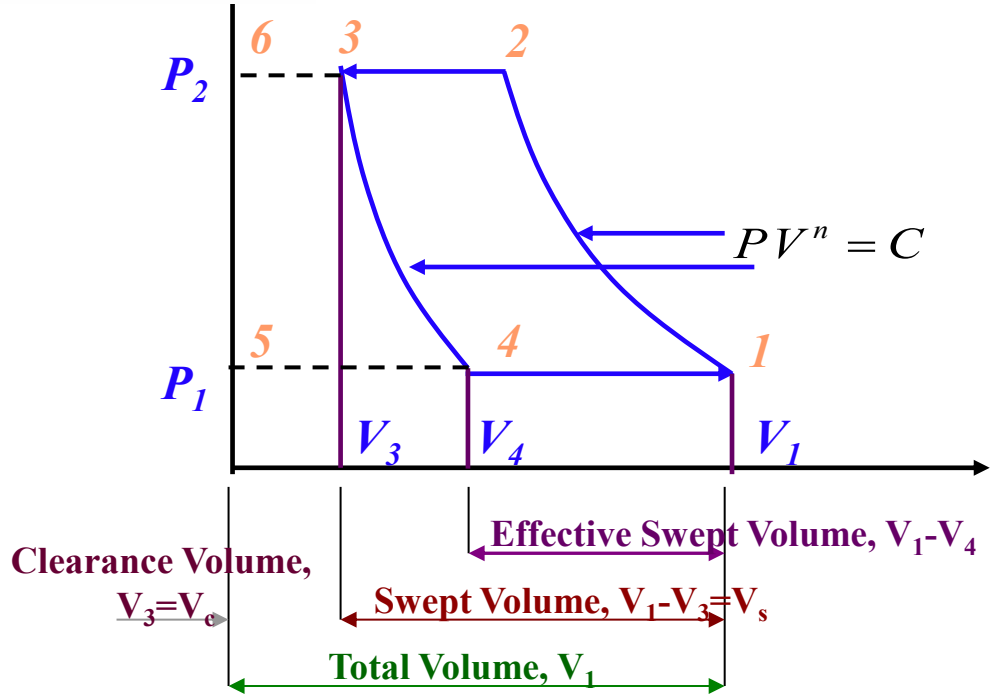
$\Rightarrow$ **1 – 2' : Isothermal Compression = Min. Work.**



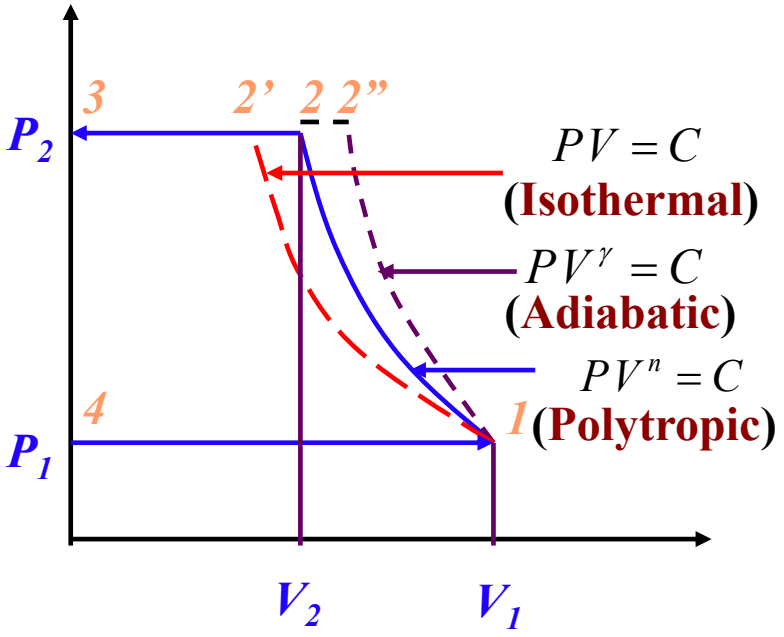
Reciprocating compressor with and without clearance







Work done without clearance volume



Actual Work = W = Area 4-1-2-3-4

W = Area (1-2) + Area (2-3)- Area (4-1)

$$= \frac{P_2V_2 - P_1V_1}{n-1} + P_2V_2 - P_1V_1$$

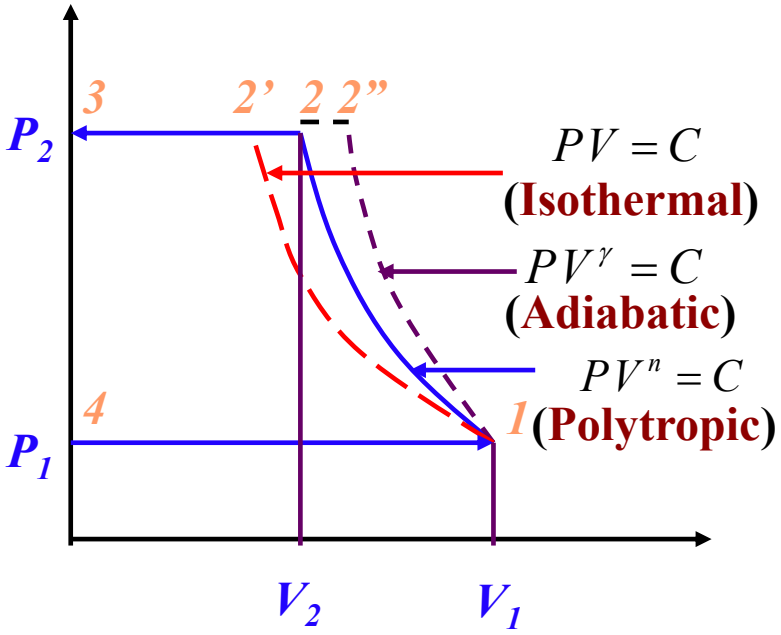
$$= (P_2V_2 - P_1V_1)\left(\frac{1}{n-1} + 1\right)$$

$$= \frac{n}{n-1} . P_1V_1\left(\frac{P_2V_2}{P_1V_1} - 1\right)$$

$$\because \frac{V_2}{V_1} = \left(\frac{P_2}{P_1}\right)^{-\frac{1}{n}} \quad [P_1V_1^{\,n} = P_2V_2^{\,n}]$$

$$W = \frac{n}{n-1} P_1V_1\left(\left(\frac{P_2}{P_1}\right)^{1-\frac{1}{n}} - 1\right)$$

From 1-2

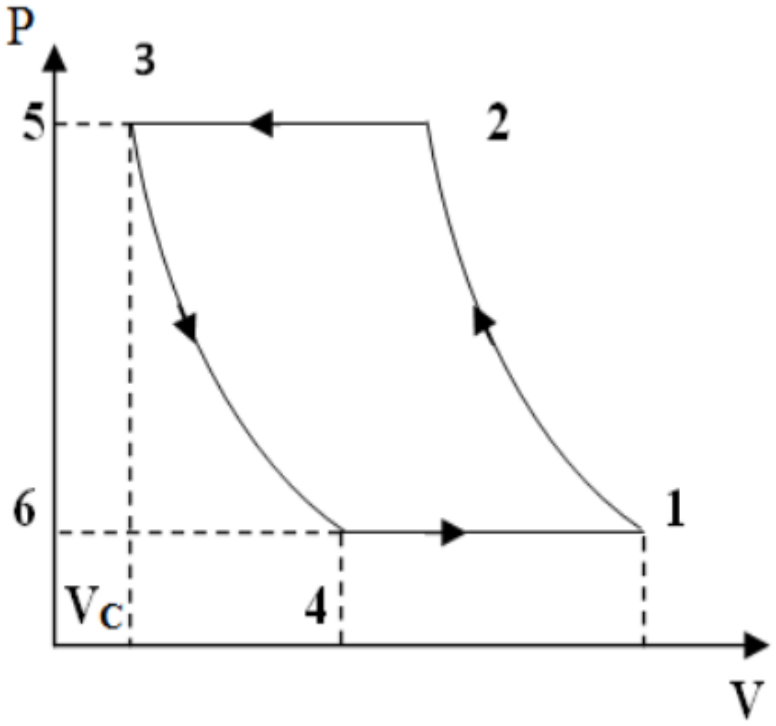


$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$$

$$= \frac{n}{n-1} p_1 v_1 \left[\frac{T_2}{T_1} - 1\right]$$

$$= \frac{n}{n-1} mRT_1 \left[\frac{T_2}{T_1} - 1\right]$$

Work done with clearance volume



Actual Work = *W* = Area 1-2-3-4-1

W **= Area (1-2-5-6-1) - Area (3-5-6-4-3)**

Work done with clearance volume

$$W = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} - 1 \right] - \frac{n}{n-1} p_4 v_4 \left[\left(\frac{p_3}{p_4}\right)^{\frac{n-1}{n}} - 1 \right]$$

But $p_4 = p_1$ & $p_3 = p_2$

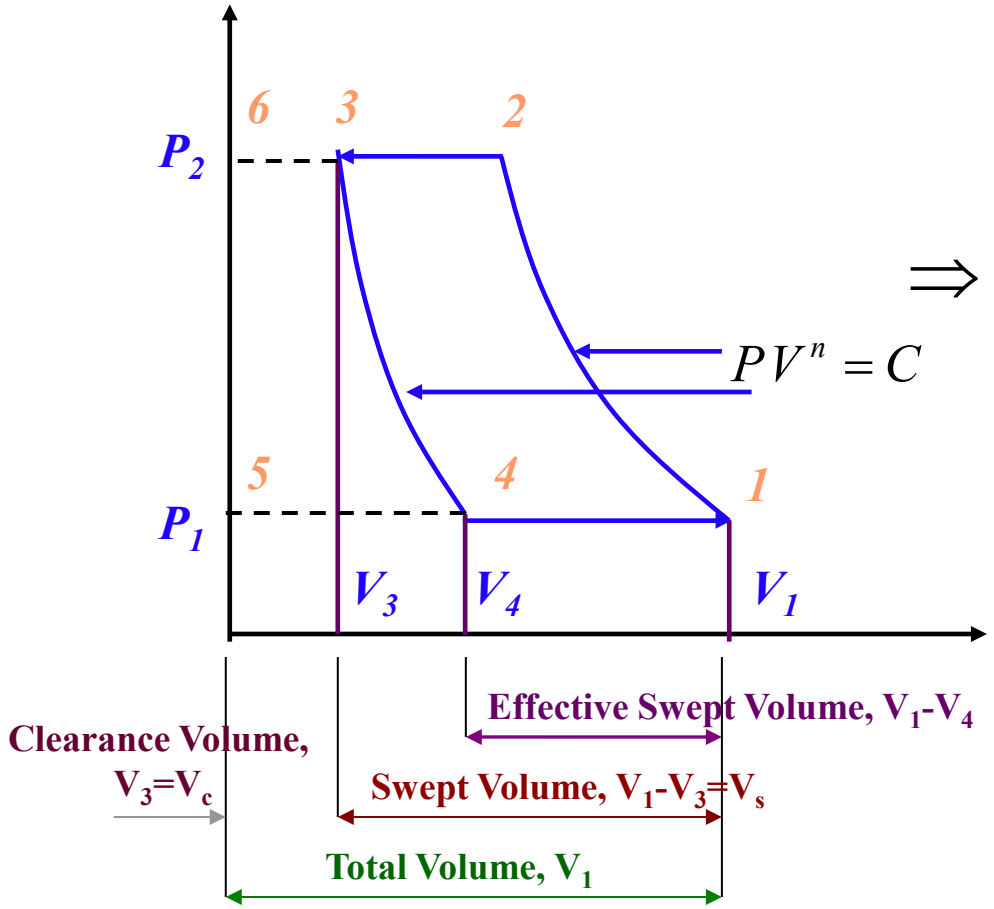
$$W = \frac{n}{n-1} p_1 v_1 \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} - 1 \right] - \frac{n}{n-1} p_1 v_4 \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} - 1 \right]$$

$$= \frac{n}{n-1} p_1 (v_1 - v_4) \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} - 1 \right]$$

$$= \frac{n}{n-1} p_1 \, (effective\ swept\ volume) \left[\left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}} - 1 \right]$$

Volumetric Efficiency

- Ratio of free air delivered to the displacement of the compressor.
- Ratio of Effective Swept Volume to Swept Volume.



$\Rightarrow$ **Presence of Clearance Volume**

Volumetric Efficiency less than 1. (60 – 85 %)

$$\text{Volumetric Efficiency} = \frac{\text{Effective Swept Volume}}{\text{Swept Volume}} = \frac{V_1 - V_4}{V_1 - V_3}$$

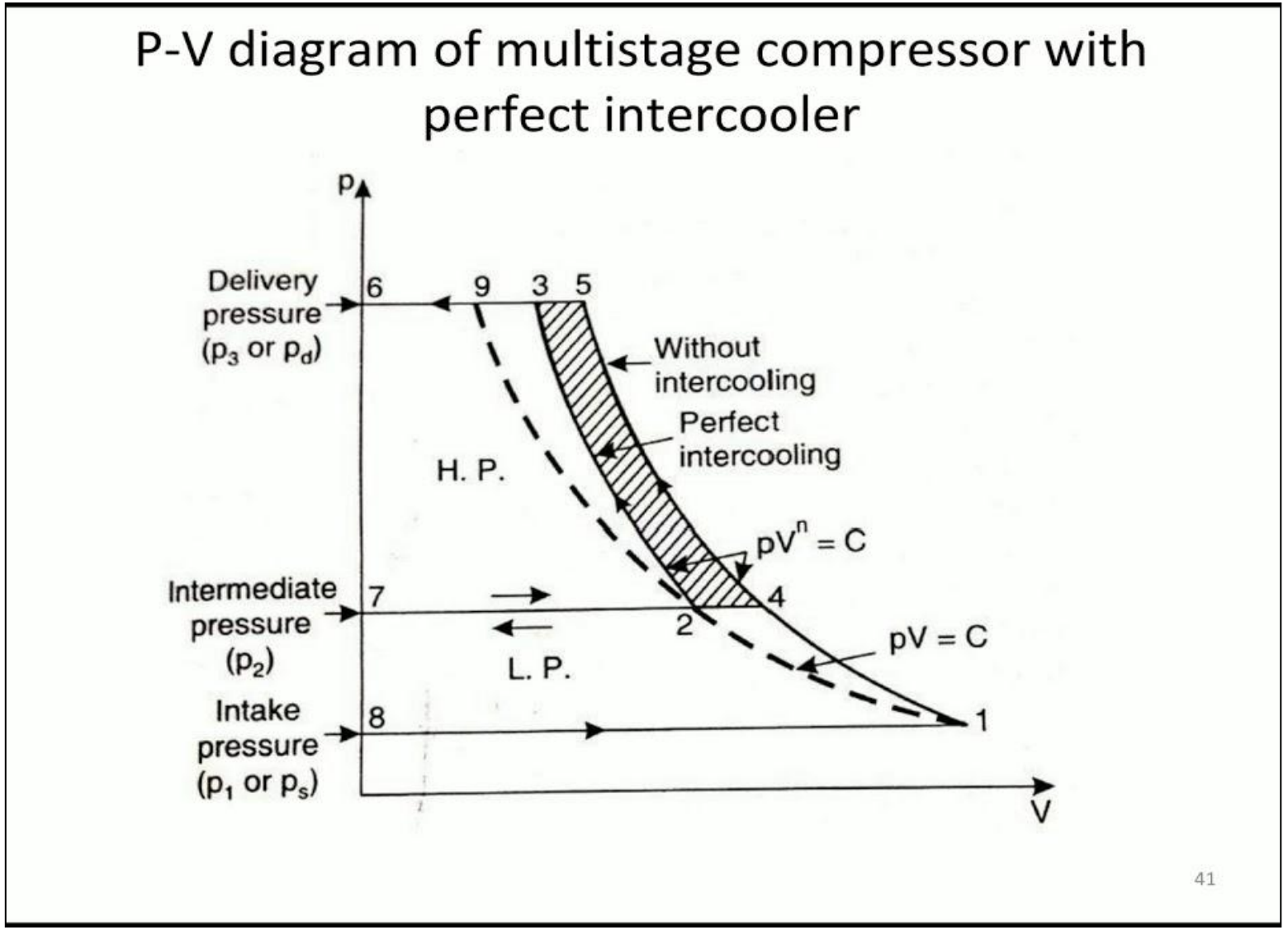
$$\text{Clearance Ratio} = \frac{\text{Clearance Volume}}{\text{Swept Volume}} = \frac{V_c}{V_s}$$

Volumetric Efficiency in terms of clearance ratio $\eta_v = 1 + K - K\left(\frac{p_2}{p_1}\right)^{1/n}$

Multistage Compressor

- In a multi stage air compressor, compression of air takes place in more than one cylinder.
- Multi stage air compressor is used in places where high pressure air is required.
- It consists of a low pressure (L.P) cylinder, an intercooler and a high pressure (H.P) cylinder. Both the pistons (in L.P and H.P cylinders) are driven by a single prime mover through a common shaft.
- The air from low pressure cylinder is then passed into an intercooler.
- In the intercooler, the air is cooled at constant pressure by circulating cold water.
- The cooled air from the intercooler is then taken into the high pressure cylinder.

Multistage Compressor



Multistage Compressor

- **Advantages:**
- ***Saving in work input:*** The air is cooled in an intercooler before entering the high pressure cylinder. Hence less power is required.
- Better balancing
- No leakage and better lubrication
- More volumetric efficiency
- High delivery pressure
- ***Cheaper materials:*** Lower operating temperature permits the use of cheaper materials for construction.

- **Disadvantages:**
- More than one cylinder is required.
- An intercooler is required. This increases initial cost. Also space required is more.
- Continuous flow of cooling water is required.
- Complicated in construction.

2-Stage Compressor (Minimum Work)

P_1 = Intake pressure (Low Pressure stage)

P_2 = Intermediate/Intercooler pressure

P_3 = Final delivery pressure (High Pressure stage)

Total Work

$$W = W_1 + W_2 = \frac{n}{n-1} P_1 V_1 \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right] + \frac{n}{n-1} P_2 V_2 \left[\left(\frac{P_3}{P_2} \right)^{\frac{n-1}{n}} - 1 \right]$$

Assumption: With perfect intercooling, the temperature entering both stages is the same, making $P_1 V_1 = P_2 V_2$.

Minimize Work: To minimize W, the derivative of W with respect to P_2 is set to zero $(\frac{dW}{dP_2} = 0)$.

Result: This optimization yields the condition that pressure ratios are equal:

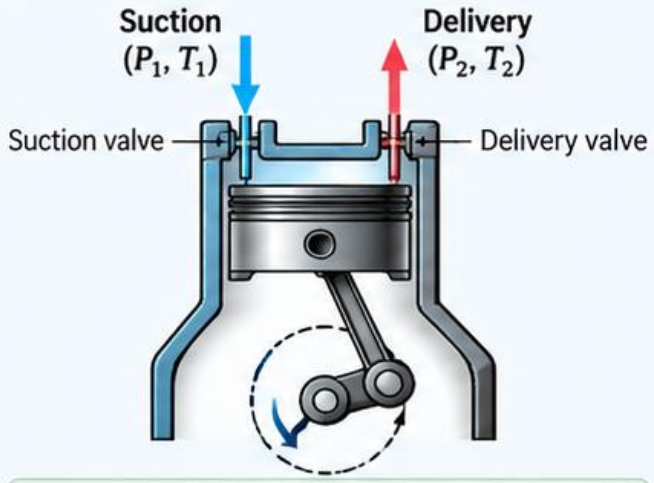
$$\frac{P_2}{P_1} = \frac{P_3}{P_2}$$

$$P_2^2 = P_1 P_3$$

$$\boldsymbol{P_2 = \sqrt{P_1 P_3}}$$

Single Stage Reciprocating Air Compressor – Key Concepts and Formulas

1. Working Principle



Single acting reciprocating compressor
(clearance may be provided)

2. Important Parameters

D	: Cylinder diameter (m)
L	: Stroke length (m)
V_s	: Swept volume (m^3)
V_c	: Clearance volume (m^3)
C	: Clearance ratio = V_c / V_s
P_1	: Suction pressure (Pa or bar)
T_1	: Suction temperature (K)
P_2	: Delivery pressure (Pa or bar)
T_2	: Delivery temperature (K)
N	: Speed of compressor (r.p.m)
n	: Polytropic index ($n = 1$ for isothermal, $n = \gamma$ for isentropic, $1 < n < \gamma$ for polytropic)
R	: Gas constant for air = 287 J/(kg·K)
η_{vol}	: Volumetric efficiency
η_{mech}	: Mechanical efficiency

3. Geometric Relations

Swept volume per stroke

$$V_s = \frac{\pi}{4} D^2 L \quad (m^3)$$

Clearance ratio

$$C = \frac{V_c}{V_s}$$

Displacement volume per minute
(theoretical at 100% volumetric efficiency)

$$\hat{V}_s = V_s N \quad (m^3/min)$$

4. Volumetric Efficiency (with clearance)

For a single stage compressor,

$$\eta_{vol} = 1 + C - C\left(\frac{P_2}{P_1}\right)^{1/n}$$

Actual volume of air inducted per minute
(at suction conditions)

$$\hat{V}_1 = \eta_{vol} V_s N \quad (m^3/min)$$

5. Mass of Air Delivered (per minute)

Based on suction conditions,

$$\rho_1 = \frac{P_1}{RT_1} \quad (kg/m^3)$$

Mass per stroke: $m_s = \rho_1 V_s = \dfrac{P_1 V_s}{RT_1}$ (kg)

Mass per minute: $\dot{m} = m_s N = \dfrac{P_1 V_s N}{RT_1}$ (kg/min)

6. Temperature after Compression

For polytropic compression,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}}$$

$$T_2 = T_1\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} \quad (K)$$

7. Work Done and Indicated Power

Polytropic work done per kg of air,

$$w = \frac{n}{n-1} RT_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

(J/kg or kJ/kg)

Indicated power,

$$IP = \dot{m} w = \frac{n}{n-1} \dot{m} RT_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

(W or kW)

8. Mechanical Efficiency and Brake Power

Mechanical efficiency, $\eta_{mech} = \dfrac{IP}{BP}$

Brake power (power required to drive the compressor),

$$BP = \frac{IP}{\eta_{mech}} \quad (W \text{ or } kW)$$

9. Free Air Delivery (FAD)

Free air delivery at standard conditions,

$$FAD = \frac{\dot{m} RT_0}{P_0} \quad (m^3/min)$$

Standard conditions
$T_0 = 273$ K
$P_0 = 1$ bar $= 10^3$ Pa

10. Speed of Compressor (from required FAD or $\dot{m}$)

For single acting compressor,

$$V_s N = \frac{\hat{V}_1}{\eta_{vol}} \quad \text{or} \quad N = \frac{\hat{V}_1}{\eta_{vol} V_s} \quad (r.p.m)$$

Numerical Problem – Reciprocating Compressor

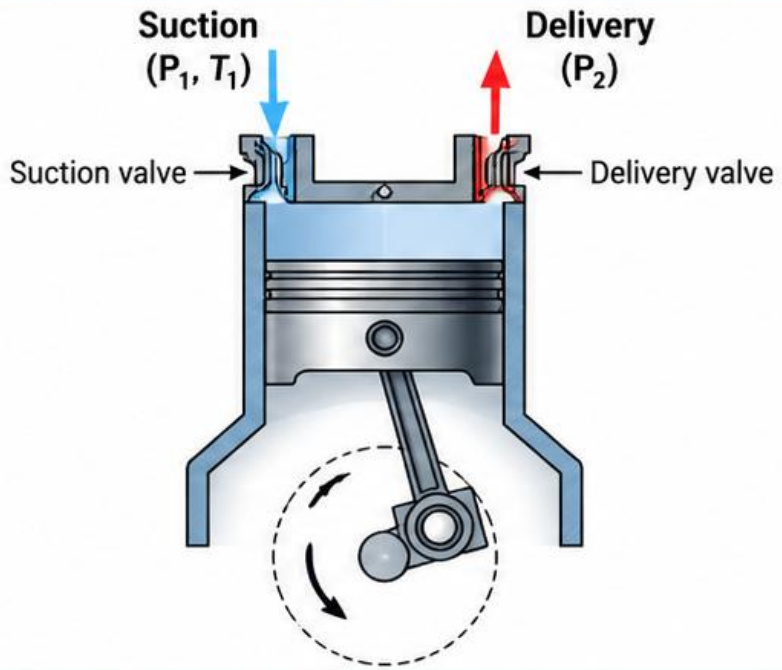
Problem Statement

A single stage reciprocating compressor takes 1 m^3 of air per minute at 1.013 bar and 15°C and delivers it at 7 bar. Assuming that the law of compression is $Pv^{1.35}$ = constant, and the clearance is negligible, calculate the indicated power.

Given Data

Suction volume, $\dot{V}_1 = 1\ m^3/min$
Suction pressure, $P_1 = 1.013\ bar = 101.3\ kPa$
Suction temperature, $T_1 = 15°C = 288.15\ K$
Delivery pressure, $P_2 = 7\ bar = 700\ kPa$
Polytropic index, $n = 1.35$
Clearance = negligible
Gas constant for air, $R = 0.287\ kJ/(kg \cdot K)$

Schematic of Single Stage Reciprocating Compressor



Single acting, single stage reciprocating compressor (clearance negligible)

1 Mass flow rate of air

Density at suction condition:

$$\rho_1 = \frac{P_1}{RT_1} = \frac{101.3}{0.287 \times 288.15} = 1.225\ kg/m^3$$

Mass flow rate:

$$\dot{m} = \rho_1 \dot{V}_1 = 1.225 \times \frac{1}{60}$$

$\dot{m}$ = 0.02042 kg/s

2 Temperature after compression

For polytropic process, $\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{(n-1)/n}$

$$T_2 = 288.15 \left(\frac{7}{1.013}\right)^{\frac{(1.35-1)}{1.35}}$$

T_2 = 475.6 K (≈ 202.5 °C)

3 Work of compression (per kg of air)

For polytropic compression ($n \neq 1$),

$$w = \frac{n}{n-1} RT_1 \left[\left(\frac{P_2}{P_1}\right)^{\left(\frac{n-1}{n}\right)} - 1\right]$$

$$w = \frac{1.35}{1.35-1}(0.287)(288.15)\left[\left(\frac{7}{1.013}\right)^{\frac{(0.35/}{1.35}} - 1\right]$$

w = 207.53 kJ/kg

4 Indicated power

Indicated power,

$$IP = \dot{m}w$$

$$IP = (0.02042)(207.53)$$

IP = 4.24 kW

Numerical Solution – Single Acting Reciprocating Air Compressor

Problem Statement

A single acting reciprocating air compressor has cylinder diameter and stroke of 200 mm and 300 mm respectively. The compressor sucks air at 1 bar and 27°C and delivers at 8 bar while running at 100 r.p.m. Find:

1. Indicated power of the compressor;
2. Mass of air delivered by the compressor per minute;
3. Temperature of the delivered air.

The compression follows the law $PV^{1.25} = C$. Take R = 287 J/kg K.

Given Data

Cylinder diameter	D	200 mm = 0.2 m
Stroke length	L	300 mm = 0.3 m
Suction pressure	P_1	1 bar = 1×10^5 Pa
Suction temperature	T_1	27°C = 300 K
Delivery pressure	P_2	8 bar = 8×10^5 Pa
Speed	N	100 r.p.m
Polytropic index	n	1.25
Gas constant for air	R	287 J/(kg K)

1 Swept Volume per Stroke

Swept volume,

$$V_s = \frac{\pi}{4} D^2 L$$

$$= \frac{\pi}{4}(0.2)^2(0.3)$$

$$= 9.425 \times 10^{-3}\ \text{m}^3$$

V_s = 9.425 × 10^{-3} m^3/stroke

2 Indicated Work per Stroke

Work done for polytropic compression,

$$W = \frac{n}{n-1} P_1 V_s \left[\left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}} - 1 \right]$$

$$W = \frac{1.25}{0.25}(1 \times 10^5)(9.425 \times 10^{-3})\left[(8)^{0.2} - 1\right]$$

$$= 2.430 \times 10^3\ \text{J/stroke}$$

W = 2.430 kJ/stroke

3 Indicated Power

For single acting compressor, working strokes per minute = N.

$$IP = \frac{WN}{60}$$

$$= \frac{2.430 \times 10^3 (100)}{60}$$

$$= 4050\ \text{W}$$

IP = 4.05 kW

4 Mass of Air Delivered per Minute

Mass of air per stroke (at suction conditions),

$$m_s = \frac{P_1 V_s}{R T_1}$$

$$m_s = \frac{(1 \times 10^5)(9.425 \times 10^{-3})}{(287)(300)}$$

$$= 0.010946\ \text{kg/stroke}$$

Mass per minute,

$$\dot{m} = m_s N$$

$$= 0.010946(100)$$

$$= \mathbf{1.095\ kg/min}$$

$\dot{m}$ = 1.095 kg/min

5 Temperature of Delivered Air

For polytropic compression,

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}}$$

$$T_2 = 300\,(8)^{0.2}$$

$$= 454.7\ \text{K}$$

T_2 = 454.7 K (181.6°C)

Numerical Problem – Single Acting Reciprocating Air Compressor

1. Problem Statement

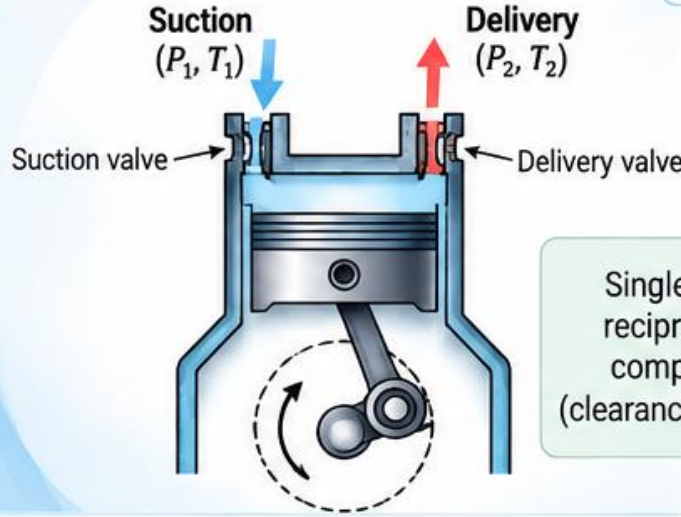
A single stage single acting air compressor delivers 0.6 kg of air per minute at 6 bar. The temperature and pressure at the end of suction stroke are 30°C and 1 bar. The bore and stroke of the compressor are 100 mm and 150 mm respectively. The clearance is 3% of the swept volume. Assuming the index of compression and expansion to be 1.3, find:

1. Volumetric efficiency of the compressor
2. Power required if the mechanical efficiency is 85%
3. Speed of the compressor (r.p.m)

2. Given Data

Mass of air delivered, n	0.6 kg/min
Delivery pressure, P_2	6 bar = 6×10^5 Pa
Suction pressure, P_1	1 bar = 1×10^5 Pa
Suction temperature, T_1	30°C = 303 K
Cylinder bore, D	100 mm = 0.1 m
Stroke length, L	150 mm = 0.15 m
Clearance volume, V_c	0.03 V_s $(C = 0.03)$
Polytropic index, n	1.3
Gas constant for air, R	287 J/(kg K)
Mechanical efficiency, η_{mech}	85% = 0.85

3. Single Acting Reciprocating Compressor



Single acting reciprocating compressor (clearance provided)

4. Volumetric Efficiency (η_{vol})

$$\eta_{vol} = 1 + C - C\left(\frac{P_2}{P_1}\right)^{1/n}$$

$$C = \frac{V_c}{V_s} = 0.03, \quad \frac{P_2}{P_1} = 6, \quad n = 1.3$$

$$\eta_{vol} = 1 + 0.03 - 0.03\,(6)^{1/1.3}$$

$$\eta_{vol} = 0.91096$$

η_{vol} = 0.911 (91.10%)

5 Indicated Power

Polytropic work of compression (per kg)

$$w = \frac{n}{n-1} RT_1 \left[\left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} - 1\right]$$

$$w = \frac{1.3}{1.3-1}(287)(303)\left[6^{\frac{1.3-1}{1.3}} - 1\right]$$

$$w = 192.97 \text{ kJ/kg}$$

Indicated power

$$\dot{m} = 0.6/60 = 0.01 \text{ kg/s}$$

$$IP = \dot{m}w = (0.01)(192.97)$$

IP = 1.93 kW

Power required (shaft power)

$$\eta_{mech} = \frac{IP}{BP}$$

$$BP = \frac{IP}{\eta_{mech}} = \frac{1.93}{0.85}$$

BP = 2.27 kW

6 Speed of the Compressor

a) Free air delivery (FAD)

$$FAD = \frac{\dot{m}RT_1}{P_1}$$

$$FAD = \frac{(0.6)(287)(303)}{1 \times 10^5}$$

FAD = 0.5218 m³/min

b) Swept volume per minute

$$\dot{V}_s = \frac{FAD}{\eta_{vol}} = \frac{0.5218}{0.91096}$$

$\dot{V}_s$ = 0.5728 m³/min

c) Speed of compressor

$$\dot{V}_s = \frac{\pi}{4} D^2 L N \quad \text{(single acting)}$$

$$0.5728 = \frac{\pi}{4}(0.1)^2(0.15)\,N$$

$$N = \frac{0.5728 \times 4}{\pi \times 0.1^2 \times 0.15}$$

N = 486.2 r.p.m

9.A two- stage acting reciprocating compressor takes in air at the rate of 0.2 m^3/s. The intake pressure and temperature of air 0.1MP_a and 16°C. The air is compressed to a final pressure of .7Mp_a. The intermediate pressure is ideal and intercooling is perfect. The compression index in both the stages is 1.25 and the compressor runs at 600 r.p.m. Neglecting clearance determine:

i. The intermediate pressure
ii. The total volume of each cylinder,
iii. The power required to drive the compressor and
iv. The rate of heat rejection in the intercooler.

Take c_p= 1.005 kJ/kg K and R =0.287 kJ/kg K

Solution .

Intake volume V_1=0.2 m^3/s
Intake pressure $p_1 = 0.1MP_a$,
Intake temperature $T_1 = 16 + 273 = 289$ K
Final pressure $p_3 = 0.7MP_a$
Compression index in both stages, $n_1 = n_2$ n =1.25
Speed of the compressor N = 600 r.p.m
c_p = 1.005kJ/kg K; R = 0.287 kJ/kg K

i. The power required to drive the compressor,P_2:

$p_2 = \sqrt{p_1 p_3} = \sqrt{0.1 \times 0.7} = 0.2646\ MP_a$

ii. The total volume of each cylinder ,V_{s1},V_{s2}:

We know that $V_{s1} \times \frac{N}{60} = V_1$ or $V_{s1} \times \frac{600}{60} = 0.2$

$\therefore$ V_{s1} (Volume of L.P cylinder) $= \dfrac{600 \times 0.2}{60} = 0.02 m_3$ (Ans).

Also $\quad p_1 V_{s1} = p_1 V_{s2}$ or $\quad V_{s2} = \dfrac{p_1 V_{s1}}{p_2}$

V_{S2} (Volume of H.P. Cylinder) $= \dfrac{0.1 \times 0.02}{0.2646} = 0.00756 m^3$ (Ans)

iii. The rate of heat rejection in the intercooler:

Mass of air handled, $m = \dfrac{p_1 V_1}{R T_1} = \dfrac{(0.1 \times 10^3) \times 0.2}{0.287 \times 289} = 0.241 kg/s$

Also, $\dfrac{T_2}{T_1} = \left(\dfrac{p_2}{p_1}\right)^{\frac{n-1}{n}}$ or $\dfrac{T_2}{289} = \left(\dfrac{0.2646}{0.1}\right)^{\frac{1.25-1}{1.25}}$ or T_2=351.1K

$\therefore$ Heat rejected in the intercooler = m $\times c_p \times (T_2 - T_1)$

= 0.241×1.005× (351.1-289) = **15.04 kJ/s or 15.04 kW.(Ans)**

Steam Nozzles

A steam nozzle is a passage of varying cross-section in which a part of the **enthalpy** of steam is converted into **kinetic energy** as the steam expands from a higher pressure to a lower pressure. Therefore, a nozzle is a device designed to **increase** the **velocity** of steam.



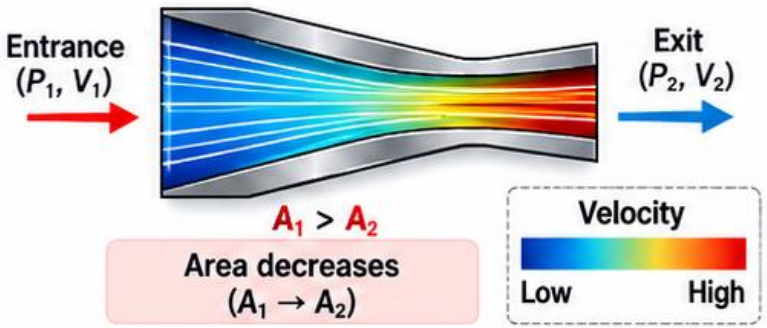
Typical steam nozzles

Types of Steam Nozzles

1. Convergent nozzle
2. Divergent nozzle
3. Convergent-divergent nozzle

1 Convergent Nozzle

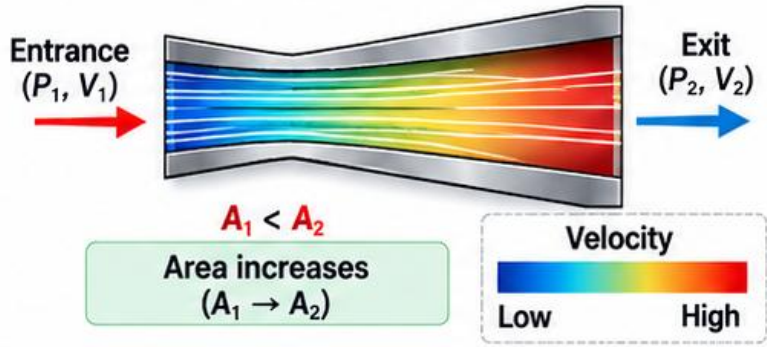
- Cross-sectional area decreases continuously from entrance to exit.
- Used when the back pressure is equal to or greater than the critical pressure ratio.
- Steam accelerates and attains higher velocity at the exit.



(a) Convergent nozzle

2 Divergent Nozzle

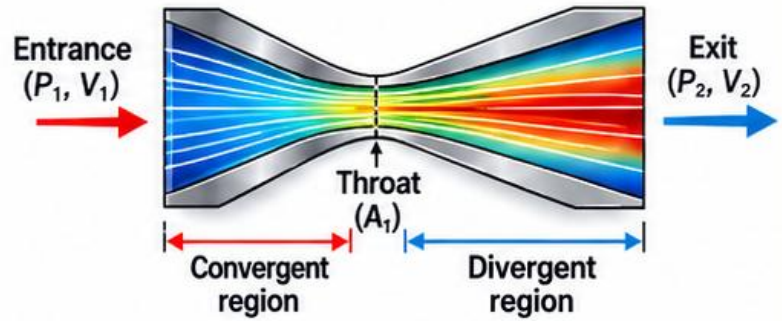
- Cross-sectional area increases continuously from entrance to exit.
- Used when the back pressure is less than the critical pressure ratio.
- Steam expands and converts pressure energy to kinetic energy (for supersonic flow).



(b) Divergent nozzle

3 Convergent-Divergent Nozzle

- Cross-sectional area first decreases from entrance to the throat and then increases from throat to the exit.
- Used when the back pressure is less than the critical pressure ratio.
- Widely used in many types of steam turbines.



(c) Convergent-divergent nozzle

Type	Variation of cross-sectional area	Condition of back pressure	Typical application
Convergent	Decreases continuously	Equal to or greater than critical pressure ratio	Low pressure ratio applications
Divergent	Increases continuously	Less than critical pressure ratio	Expansion to supersonic velocities
Convergent-divergent	Decreases to throat, then increases	Less than critical pressure ratio	Steam turbines and high pressure ratio applications

Steam Nozzle – Principle, Flow Behaviour and Governing Equations

1 Introduction

- A steam nozzle is a passage of varying cross-section in which part of the enthalpy of steam is converted into kinetic energy as the steam expands from a higher pressure to a lower pressure.
- Used to accelerate steam in steam turbines.
- The same principle is used in jet and rocket engines.
- Ideally, no external work is done; the decrease in enthalpy appears as an increase in velocity, while pressure and temperature decrease.
- Common names: Convergent nozzle, Divergent nozzle, Convergent–Divergent (C–D) nozzle, de Laval nozzle.

2 Governing Equations (Steady, Adiabatic Flow)

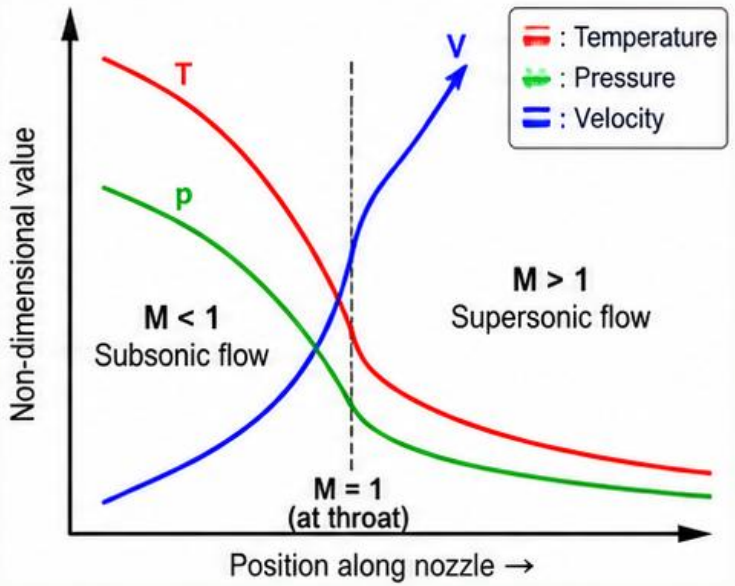
1. **Continuity equation** $\dot{m} = \rho A V = \text{constant}$
2. **Steady flow energy equation** (no work, negligible potential energy)

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} = h_0$$

3. **T–ds relationship** $T\,ds = dh - \frac{dp}{\rho}$
4. **Isentropic relationships (ideal gas)**

$$\frac{p}{\rho^{\gamma}} = \text{constant}, \quad \frac{T}{p^{(\gamma-1)/\gamma}} = \text{constant}$$

3 Variation of Properties in a Nozzle



4 Types of Steam Nozzles

(a) Convergent Nozzle



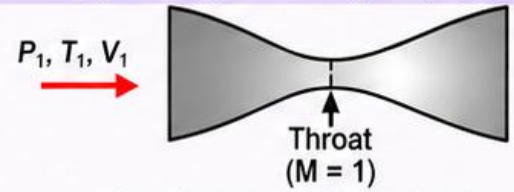
(Area decreases continuously)

(b) Divergent Nozzle



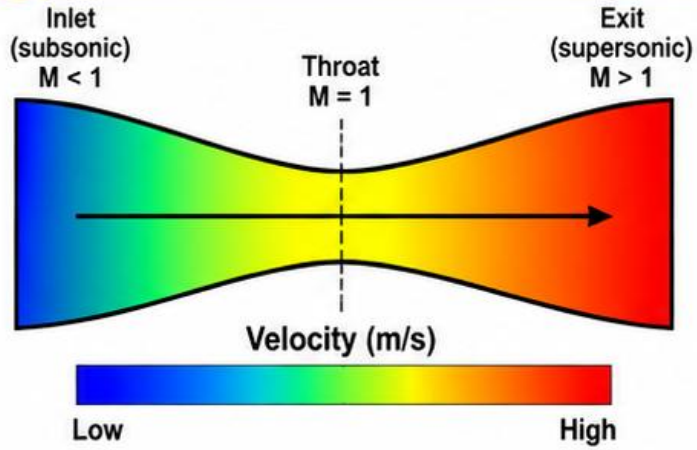
(Area increases continuously)

(c) Convergent–Divergent (C–D) Nozzle



(Area decreases to throat, then increases)

5 Flow through a Convergent–Divergent Nozzle



6 Actual Steam Turbine Nozzle



Construction and Working

- Why should area increase for accelerating a supersonic flow?

$$\frac{dA}{A} = -\frac{dV}{V}\left(1 - M^2\right)$$

- The change in area (+ or -) depends on the Mach number
- Acceleration requires a converging passage for subsonic flows (M < 1) and diverging passage for supersonic (M > 1)
- The nozzle receives steam from a high pressure chamber (e.g. a boiler). Pressure = p_c ; Temperature = T_c; Velocity, $V_c \sim 0$
- The section of minimum area is called 'throat': p_t, T_t, V_t
- Discharges a jet at exit: back pressure p_b
- Back pressure regulated by a flow valve : Less p_b gives more flow
- <u>One dimensional flow analysis:</u> Flow properties are constant across any cross section, no boundary layer effects

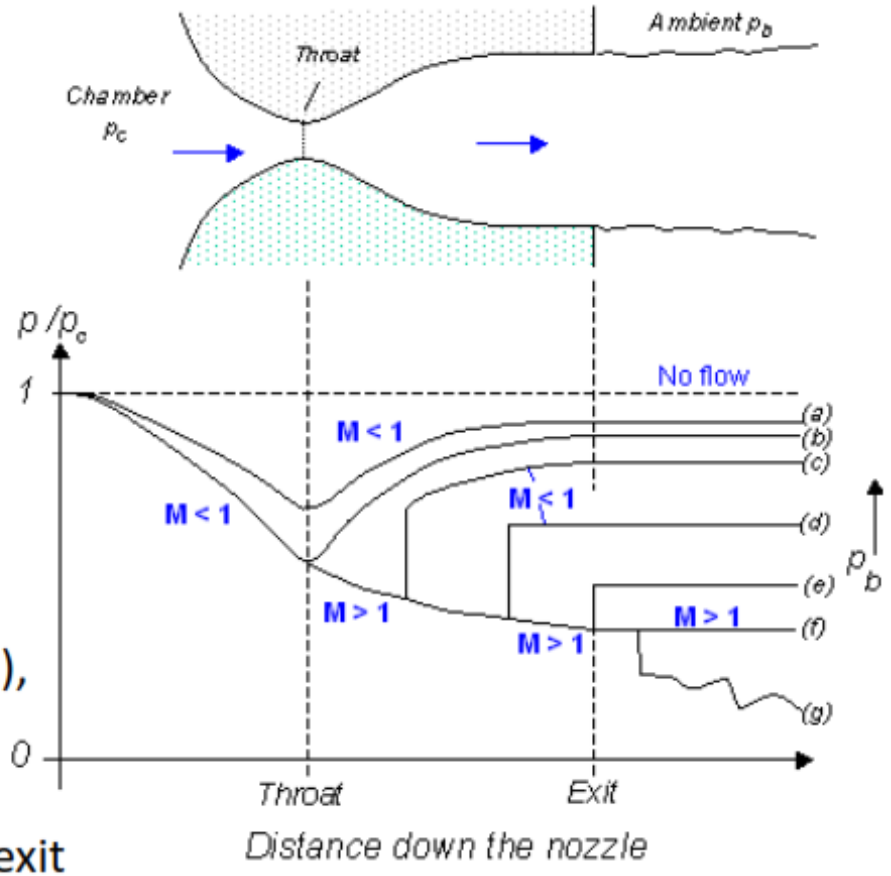
Effect of Back Pressure on Flow

- Flow is controlled by the pressures
- Greater Δp, more acceleration
- Chamber pressure is fixed, back pressure reduced progressively (by means of a flow valve etc.)
- No flow when $p_b = p_c$

a) p_b slighly < p_c ; Flow accelerates till throat (M<1), then decelerates

b) Flow sonic at throat (M=1), flow is choked, decelerates downstream

c) Flow accelerates beyond throat (M >1), suddenly decelerates across a normal shock wave, sudden pressure rise



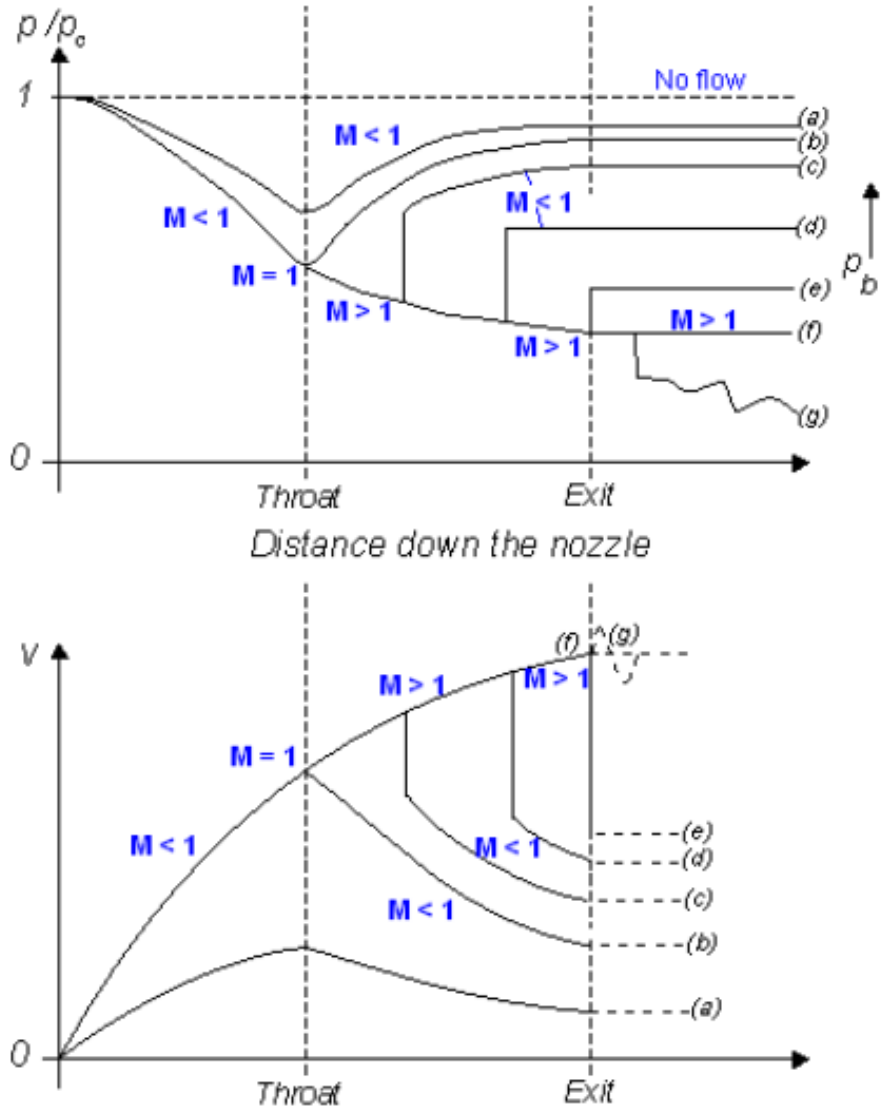
d) Shock shifts downstream, e) shock at exit

f) Design condition, no shock, M> 1 throughout divergent portion

g) Underexpanded, pressure reduces across expansion wave at exit

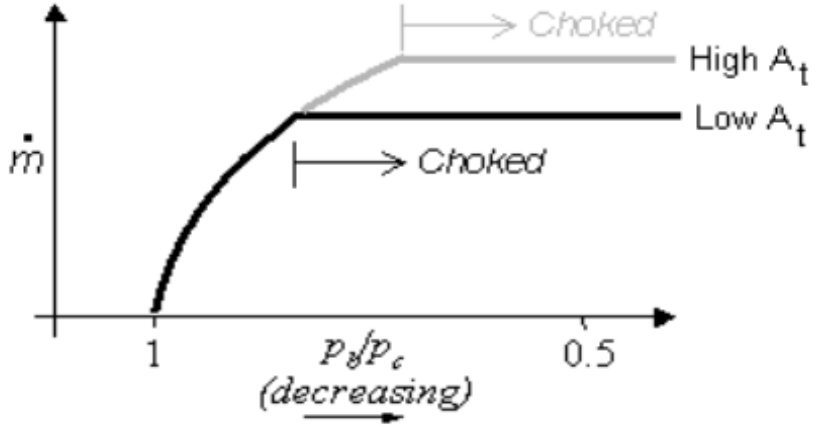
Variation in Velocity in C-D Nozzle

- Velocity at throat is same for conditions b-g : Choked flow
- T, p, ρ at throat are also same for b-g (from SFEE, isentropic relationships)
- Hence mass flow rates (= ρAV) are also equal for conditions b-g
- Note that the drop in velocity across shock increases as shock location shifts downstream (c – e)
- This is because temperatures and thus sonic velocity are lower at downstream points



Points to Note

- Pressure, Temperature drop during acceleration (irrespective of whether subsonic/supersonic)
- Once the velocity at throat becomes sonic, the flow becomes choked. Further fall in p_b cannot increase the mass flow rate. The flow pattern in converging section also remains constant

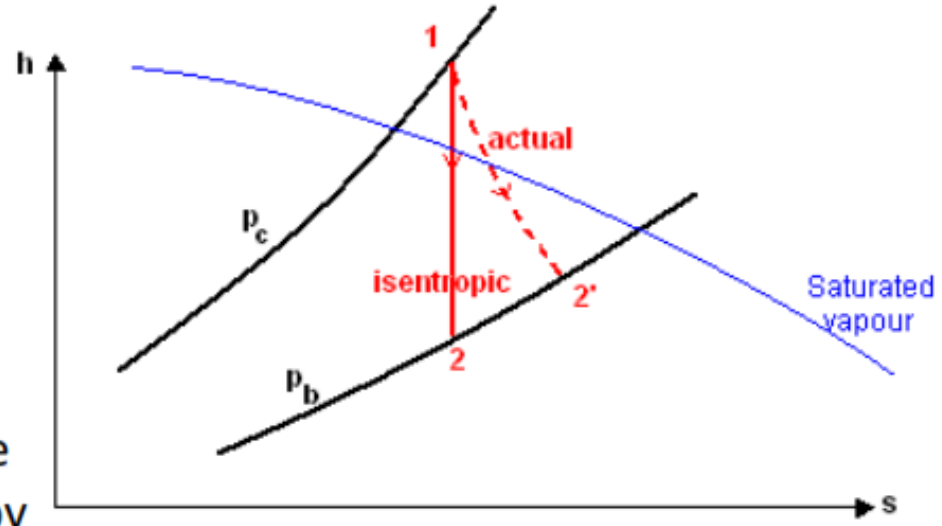


- Flow rate can be increased by widening throat area (A_t), but choking would still occur at some lower value of back pressure
- A normal shock wave is necessary to convert a supersonic flow to subsonic in the divergent passage
- p, T rise sharply (instantaneously) across a shock wave - irreversible
- An expansion wave causes a reduction in pressure

If the mass flow rate is reached the maximum value at the throat then the flow is called choked flow

Actual and Ideal Operation of Nozzle

- For a given Δp, highest reduction in enthalpy is obtained for isentropic expansion (1-2)
- Hence isentropic expansion gives maximum acceleration
- Actual expansion (1-2'), though still adiabatic, has irreversible effects like friction, shock etc. Generates entropy
- Actual Δh, ΔV are lower
- Isentropic efficiency compares actual Δh to isentropic Δh
- Most of friction occurs in divergent passage (why?), has less effect on flow rate (determined by choking)
- Friction reduces KE, increases dryness fraction and specific volume at exit



$$\eta_s = \frac{\Delta h_{act}}{\Delta h_{isentropic}} = \frac{h_1 - h_{2'}}{h_1 - h_2}$$

$$\eta_s = \frac{\Delta KE_{act}}{\Delta KE_{isentropic}} = \frac{V_{2'}^2 - V_1^2}{V_2^2 - V_1^2} \cong \frac{V_{2'}^2}{V_2^2}$$

CD Nozzle – Important Equations

- Velocity at any section 2:

$$V_2 = \sqrt{2(h_1 - h_2)} = \sqrt{\frac{2\gamma}{\gamma - 1}\left(\frac{p_1}{\rho_1} - \frac{p_2}{\rho_2}\right)}$$

$$= \sqrt{\frac{2\gamma}{\gamma - 1}\frac{p_1}{\rho_1}\left(1 - \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}\right)} = a_1\sqrt{\frac{2}{\gamma - 1}\left(1 - \frac{T_2}{T_1}\right)}$$

- Mach no. at any section 2: $M_2 = \sqrt{\frac{2}{\gamma - 1}\left(\frac{T_1}{T_2} - 1\right)} = \sqrt{\frac{2}{\gamma - 1}\left(\left(\frac{p_1}{p_2}\right)^{\frac{\gamma-1}{\gamma}} - 1\right)}$

- Max velocity: $V_{max} = a_1\sqrt{\frac{2}{\gamma - 1}}$

- Critical pressure, velocity (at throat, M = 1): $p^* = p_1\left(\frac{2}{\gamma + 1}\right)^{\frac{\gamma}{\gamma-1}}$; $V^* = a_1\sqrt{\frac{2}{\gamma + 1}}$

Equations in red boxes are general. Others are special cases for ideal gases, Inlet velocity is neglected : $V_1 \sim 0$

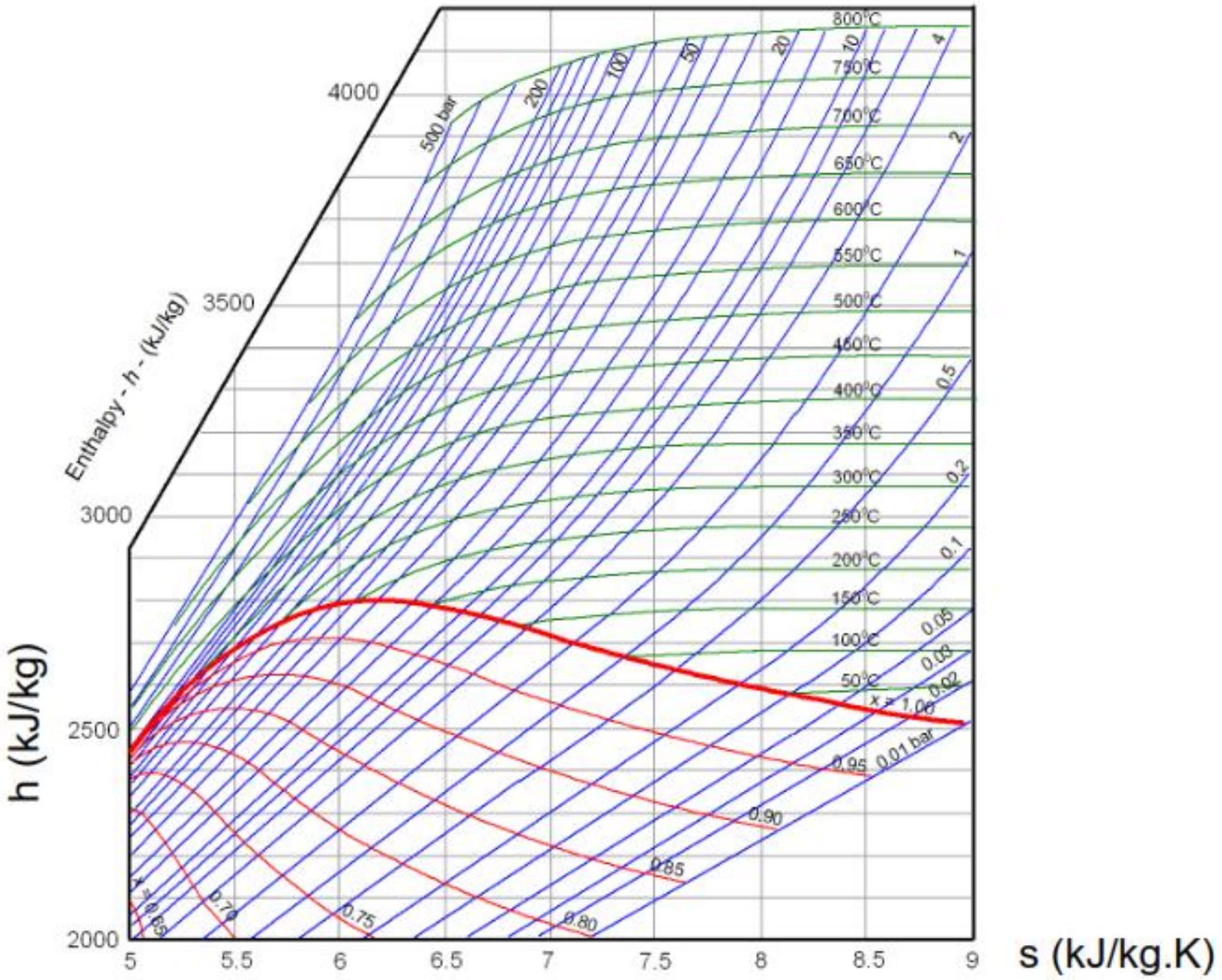
Estimating Properties of Steam

- Steam tables
  - Saturated (w.r.t T_{sat} and p_{sat})
  - Superheated (w.r.t different T for a given p_{sat})
- Can be downloaded from: http://enpub.fulton.asu.edu/ece340/pdf/steam_tables.PDF
- Online calculator: http://www.peacesoftware.de/einigewerte/wasser_dampf_e.html

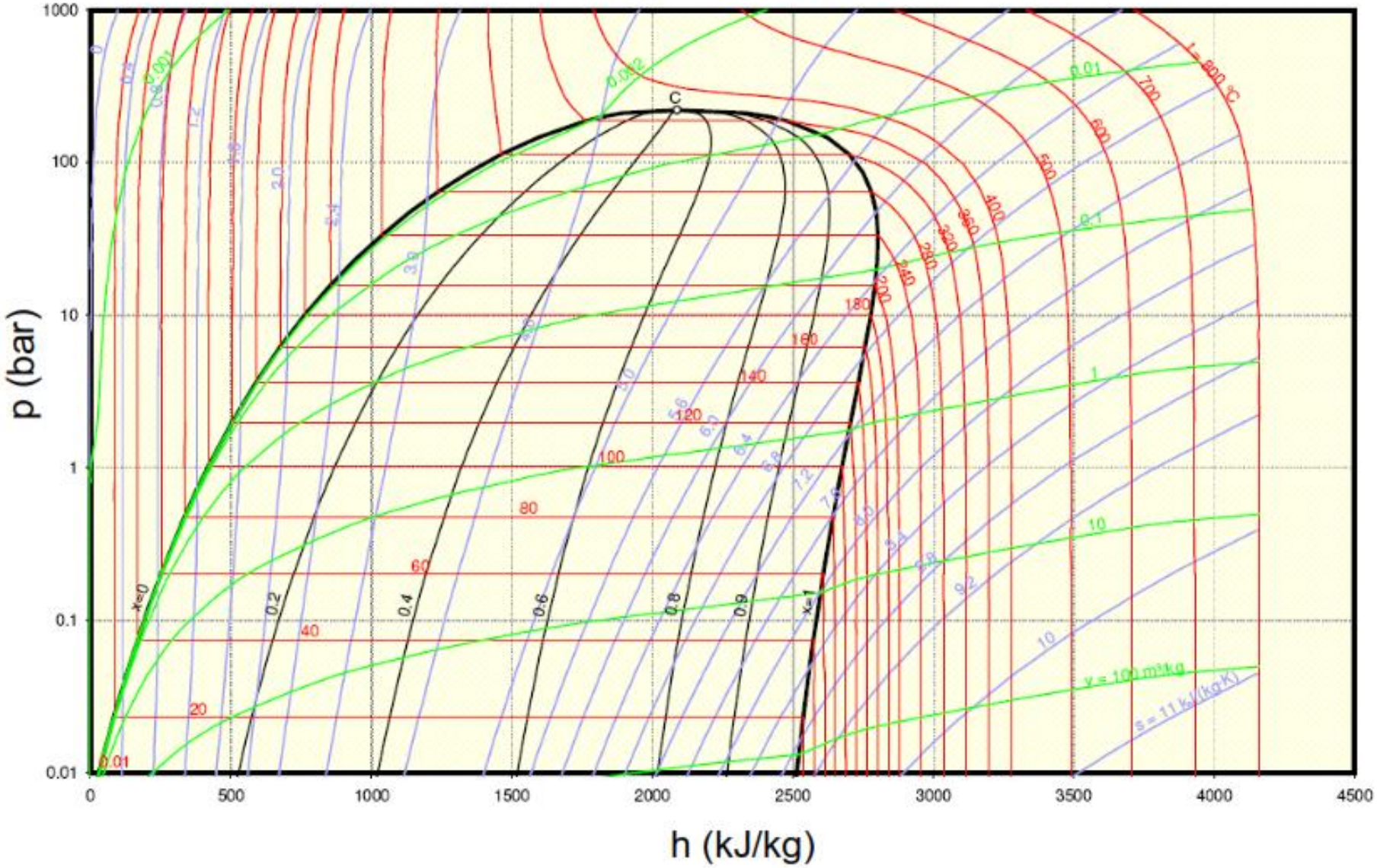
p [bar]	T_s [°C]	v_g [m³/kg]	u_f [kJ/kg]	u_g [kJ/kg]	h_f [kJ/kg]	h_{fg} [kJ/kg]	h_g [kJ/kg]	s_f [kJ/kg K]	s_{fg} [kJ/kg K]	s_g [kJ/kg K]
0.006112	0.01	206.1	0†	2375	0*	2501	2501	0†	9.155	9.155
0.010	7.0	129.2	29	2385	29	2485	2514	0.106	8.868	8.974
0.015	13.0	87.98	55	2393	55	2470	2525	0.196	8.631	8.827
0.020	17.5	67.01	73	2399	73	2460	2533	0.261	8.462	8.723
0.025	21.1	54.26	88	2403	88	2451	2539	0.312	8.330	8.642
0.030	24.1	45.67	101	2408	101	2444	2545	0.354	8.222	8.576
0.035	26.7	39.48	112	2412	112	2438	2550	0.391	8.130	8.521
0.040	29.0	34.80	121	2415	121	2433	2554	0.422	8.051	8.473
0.045	31.0	31.14	130	2418	130	2428	2558	0.451	7.980	8.431
0.050	32.9	28.20	138	2420	138	2423	2561	0.476	7.918	8.394

T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg·K
P = 8.0 MPa (T_{SAT}=295.06° C)				
Sat.	0.0235	2569.8	2758.0	5.7432
320	0.0268	2662.7	2877.2	5.9489
360	0.0309	2772.7	3019.8	6.1819
400	0.0343	2863.8	3138.3	6.3634
440	0.0374	2946.7	3246.1	6.5190
480	0.0403	3025.7	3348.4	6.6586
520	0.0431	3102.7	3447.7	6.7871
560	0.0458	3178.7	3545.3	6.9072
600	0.0485	3254.4	3642.0	7.0206
640	0.0510	3330.1	3738.3	7.1283
700	0.0548	3443.9	3882.4	7.2812
740	0.0573	3520.4	3978.7	7.3782

h-s Diagram/Chart [Mollier Diagram]



p-h Diagram/Chart



Critical Pressure Ratio (P2/P1)

There is only one value of the ratio (p_2 / p_1) which produces the maximum discharge from the nozzle. This ratio is called *critical pressure ratio.*

where p_1 be the inlet pressure

p_2 be the throat pressure

(i) For saturated steam $n = 1.135$

The critical pressure ratio is given by

$$\frac{p_2}{p_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

$$\frac{p_2}{p_1} = \left(\frac{2}{1.135+1}\right)^{\frac{1.135}{1.135-1}}$$

Critical pressure ratio, $\frac{p_2}{p_1} = 0.577$

(ii) For superheated steam n = 1.3

$$\frac{p_2}{p_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

$$\frac{p_2}{p_1} = \left(\frac{2}{1.3-1}\right)^{\frac{1.3}{1.3-1}}$$

Critical pressure ratio, $\frac{p_2}{p_1} = 0.546$

(iii) For gases n = 1.4

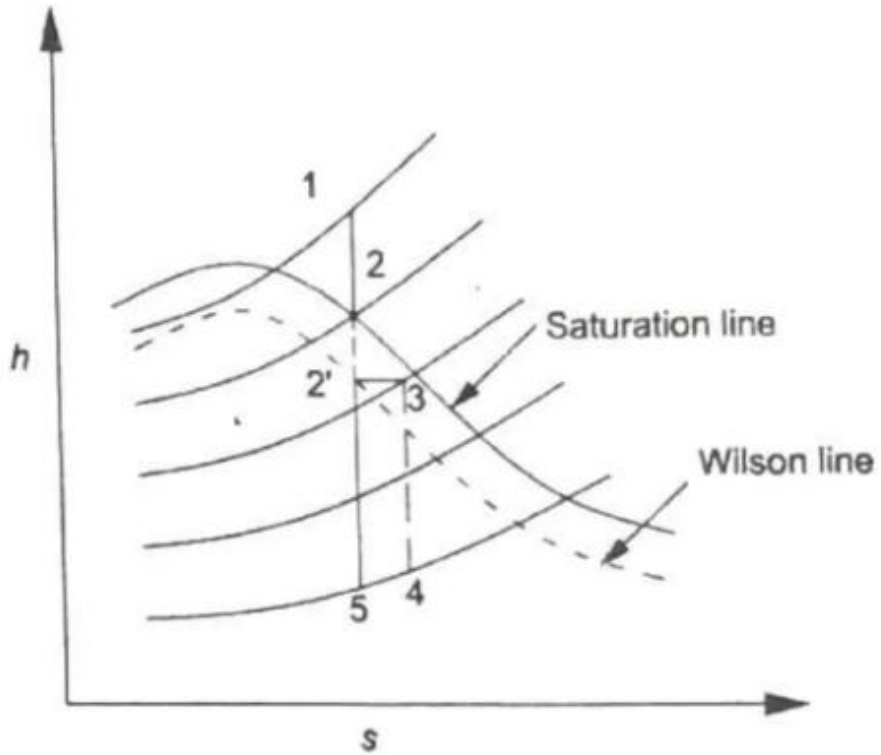
Critical pressure ratio, $\frac{p_2}{p_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$

$$\frac{p_2}{p_1} = \left(\frac{2}{1.4+1}\right)^{\frac{1.4}{1.4-1}}$$

$$\frac{p_2}{p_1} = 0.5282$$

Metastable Flow (or) Supersaturated Flow

When the superheated steam expands in a nozzle, the condensation occurs in the nozzle. Since the steam has more velocity, the condensation will not take place at the expected rate. So, the equilibrium between liquid and vapour phase is delayed and the steam continues to expand in a dry state. The steam in such a set of conditions is said to be *supersaturated* or *metastable flow*



The vapour between pressures p_2 and p_3 is said to be supersaturated and this type of flow in the nozzle is known as *supersaturated* or *metastable flow of steam*.

Wilson Line

A limit to the supersaturated state was observed by Wilson and a line drawn on the chart through the observed point is known as *Wilson line*.

Effects of Supersaturation

The following are the effects of supersaturation in flow of steam through a nozzle:

1. The dryness fraction of the steam is increased.
2. Entropy and specific volume of the steam are increased.
3. The exit velocity of the steam is reduced.
4. Mass of steam discharged is increased.

Dry saturated steam at a pressure of 11 bar enters a convergent-divergent nozzle and leaves at a pressure of 2 bar. The flow is adiabatic and frictionless and neglects the inlet velocity of steam. If the isentropic specific enthalpy drop between inlet and exit is 180 kJ/kg, what will be the exit velocity of steam?

The nozzle is a duct of varying cross-sectional area in which velocity increase with a corresponding drop in pressure.

The steady flow energy equation for a nozzle can be written as,

$$h_1 + \frac{C_1^2}{2000} + Q = h_2 + \frac{C_2^2}{2000} + W$$

And for adiabatic and frictionless flow, Q = 0

and also there is no shaft work in the nozzle, so, W = 0

h_1 = Specific enthalpy at the inlet of nozzle,

h_2 = Specific enthalpy at the outlet of nozzle,

C_1, and C_2 are velocities at the inlet and outlet of the nozzle respectively.

For nozzle, SFEE can be written as,

$$h_1 + \frac{C_1^2}{2000} = h_2 + \frac{C_2^2}{2000}$$

Or velocity at the outlet of the nozzle can be written as,

$$C_2 = \sqrt{C_1^2 + 2000(h_1 - h_2)}$$

Calculation:

Given:

Pressure at inlet, p_1 = 11 bar

Pressure at outlet, p_2 = 2 bar,

Inlet velocity is neglected, i.e., $C_1 = 0$,

Isentropic enthalpy drop, $h_1 - h_2$ = 180 kJ/kg

To find -

Outlet velocity, C_2 = ?

So now on putting the given values in the formula,

$$C_2 = \sqrt{2000(h_1 - h_2)}$$

$$C_2 = \sqrt{2000 \times 180} = 600\ m/s$$

So, exit velocity of steam, C_2 = 600 m/s

Numerical - Steam Nozzles

At design conditions, dry saturated steam at 10 bar is expanded in a nozzle to 0.4 bar. The throat area is 7 cm^2 and the inlet velocity is negligible. Estimate the mass flow and the exit area. Assume isentropic flow and take the index of expansion n = 1.135 for dry saturated steam.

p_1 = 10 bar; p_2 = 0.4 bar; A_t = 7 x 10^{-4} m^2; n = 1.135

For saturated condition at inlet, $T_1 = T_{sat,g}$ @ 10 bar = 179.9 °C,

Also, $v_1 = v_g$ = 0.194 m^3/kg; $h_1 = h_g$ = 2778.1 kJ/kg; $s_1 = s_g$ = 6.587 kJ/kg.K

At design conditions, $p_t = p^* = p_1 \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}} = 5.8 \text{ bar}$

As expansion is isentropic, at throat, $s_t = s_1$ = 6.587 kJ/kg.K

Also, $s_t = s_f + x.s_{fg}$

At 5.8 bar, s_f = 1.918 kJ/kg.K and s_{fg} = 4.853 kJ/kg.K

from which, dryness fraction x_t = 0.9621

Numerical - Steam Nozzles [Contd.]

Similarly, enthalpy a throat, $h_t = h_f + x.h_{fg} = 664.8 + 0.9621*2090.5 = 2676.1$ kJ/kg

Neglecting inlet velocity, velocity at throat, $V_t = V^* = \sqrt{2(h_1 - h_t)} = 451.7$ m/s

Specific volume at throat, $v_t \approx x.v_g$ (why are we not taking v_f into account?)

$= 0.3136$ m^3/kg

Hence, mass flow rate, $\dot{m} = \frac{A_t V_t}{v_t} = 1.008$ kg/s

Proceed in a similar manner for flow at exit:

$x_2 = 0.837$; $h_2 = 2258.8$ kJ/kg; $V_2 = 1019.4$ m/s, $v_2 = 3.34$ m^3/kg; $A_2 = 3.3 \times 10^{-3}$ m^2

Extra: What will be the exit conditions, if isentropic efficiency is 94%?

Expansion follows pv^n = constant

n = 1.135 for saturated steam,

n = 1.3 for superheated steam

In general, n = 1.035 + 0.1x [Zenner's equation]

where x is the initial dryness fraction

Numerical

- Steam is expanded in set of nozzles from 10 bar and 200 ° C to 5 bar. What type of nozzle is it? Neglecting the initial velocity find minimum area of the nozzle required to allow a flow of 3 kg/s under the given conditions. Assume that expansion of steam to be isentropic.

Given:

Pressure at the entry to the steam nozzles

$P_1 = 10$ bar

$T_1 = 200^\circ C$

Steam at exit pressure

$P_2 = 5$ bar

We know that

$$\frac{P_2}{P_1} = \left(\frac{2}{n+1}\right)^{\frac{n}{n-1}}$$

$$= \left(\frac{2}{1.3+1}\right)^{\frac{1.3}{0.3}}$$

$$= \left(\frac{2}{2.3}\right)^{4.333} = 0.5457$$

$$P_2 = P_1 \times 0.5457$$

$$= 10 \times 0.5457$$

$$P_2 = 5.5 \text{ bar}$$

Since throat pressure (P_2) is greater than the exit pressure the nozzle used is Convergent divergent nozzle. The minimum area will be at throat, where the pressure is 5.5 bar.

Pressure at 10 bar h_1 =

s_1

h_1 = s_1

$s_1 = s_2$

$s_1 = s_{f_2} + x_2 s_{fg_2}$

$h_2 = h_{f_2} + x_2 h_{fg_2}$

$h_2 = h_{f_2} + x_2 (h_{g_2} - h_{f_2})$

Specific volume of 5.5 bar

$v = 0.345 \ m^3/kg.$

Velocity of Throat $V_2 = 44.72 \sqrt{h_1 - h_2}$

$= 44.72 \sqrt{\quad}$

Throat area $A_2 = \frac{mv}{\rho_2 V_2}$

$= \frac{3 \times 0.345}{\quad}$

$A = 0.0021 \ m^2$

6. ***Dry saturated steam enters a nozzle at a pressure of 10 bar and with an initial velocity of 90 m/s. The outlet pressure is 6 bar and the outlet velocity is 435 m/s. The hear loss from the nozzle id 9 kJ/kg of stream flow.***

Calculate the dryness fraction and the area at the exit, if the area at the inlet is 1256 mm^2

Given:

p_1 =10 bar; V_1 = 90 m/s;

p_3= 6 bar; V_3= 435 m/s;

Loses = 9 kJ/Kg;

A_1=1256 mm^2 = 1256 x 10^{-6} m^2

Solution.

Dryness fraction of steam

Let x_3 = Dryness fraction of steam at the exit.

From steam tables,

corresponding to a pressure of 10 bar, we find that enthalpy of dry saturated steam,

h_1 = 2776.2 kJ/kg; and $v_{g1} = 0.1943 m^3/kg$ and

corresponding to a pressure of 6 bar, we find that

$$h_{\beta} = 670.4kJ/kg;\ h_{fg3} = 2085kK/kg;\ v_{g3} = 0.3155m^3/kg$$

We know that for a steady flow through the nozzle,

$$h_1 + \frac{1}{1000}\left(\frac{V_1^2}{2}\right) = h_3 + \frac{1}{1000}\left(\frac{V_3^2}{2}\right) + Losses$$

$$\therefore \quad h_3 = h_1 + \frac{2}{1000}(V_1^2 - V3^2) - Losses$$

$$= 2776.2 + \frac{1}{2000}[(90)^2 - (435)^2] - 9$$

$$= 2776.2\text{-}99.6$$

$$= 2676.6 \text{ kJ/kg}$$

We also know that enthalpy of wet stream (h_3),

$$2676.6 = h_\beta + x_3 h_{fg3} = 670.4 + x_3 \times 2085$$

$$\therefore \qquad x_3 = 0.962 \text{ Ans.}$$

Area at exit

Let A_3 Area at exit in m^2

We know that $\frac{A_1 V_1}{x_1 v_{g1}} = \frac{A_3 V_{31}}{x_{31} v_{g3}}$ *or* $\frac{1256x10^{-6} \times 90}{1 \times 0.1943} = \frac{A_3 \times 435}{0.962 \times 0.3155}$

…(For dry saturated steam, $x_1 = 1$)

$A_3 = 406 \text{ x } 10^{-6} m^2 = 406 \text{ mm}^2$ Ans.